



OIL INDUSTRY SAFETY DIRECTORATE

पेट्रोलियम डिपो, टर्मिनलों, पाईप लाइन प्रतिष्ठानों
और ल्यूब ऑइल प्रतिष्ठानों की अग्नि संरक्षण सुविधाएं
ओ आई एस डी-मानक -117

**FIRE PROTECTION FACILITIES
FOR PETROLEUM DEPOTS, TERMINALS, PIPELINE
INSTALLATIONS AND LUBE OIL INSTALLATIONS**

OISD-STANDARD-117

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OIL INDUSTRY SAFETY DIRECTORATE

Government of India

Ministry of Petroleum & Natural Gas

8th Floor, OISD Bhavan, Plot No. 2, Sector – 73, Noida – 201301 (U.P.)

Website: <https://www.oisd.gov.in>

Tele: 0120-2593833

PREAMBLE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for sovereignty and prosperity of the country. As the industry essentially deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as only safe performance at all times can ensure optimum ROI of these national assets and resources including sustainability.

While statutory organizations were in place all along to oversee safety aspects of Indian petroleum industry, Oil Industry Safety Directorate (OISD) was set up in 1986 by Ministry of Petroleum and Natural Gas, Government of India as a knowledge centre for formulation of constantly updated world-scale standards for design, layout and operation of various equipment, facility and activities involved in this industry. Moreover, OISD was also given responsibility of monitoring implementation status of these standards through safety audits.

In more than three decades of its existence, OISD has developed a rigorous, multi-layer, iterative and participative process of development of standards – starting with research by in-house experts and iterating through seeking & validating inputs from all stake-holders – operators, designers, national level knowledge authorities and public at large – with a feedback loop of constant updation based on ground level experience obtained through audits, incident analysis and environment scanning.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry culminating in a safer environment in the industry. OISD – except in the Upstream Petroleum Sector – is still a regulatory (and not a statutory) body but that has not affected implementation of the OISD standards. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption in various statutory rules of the land.

Petroleum industry in India is significantly globalized at present in terms of technology content requiring its operation to keep pace with the relevant world scale standards & practices. This matches the OISD philosophy of continuous improvement keeping pace with the global developments in its target environment. To this end, OISD keeps track of changes through participation as member in large number of International and national level Knowledge Organizations – both in the field of standard development and implementation & monitoring in addition to updation of internal knowledge base through continuous research and application surveillance, thereby ensuring that this OISD Standard, along with all other extant ones, remains relevant, updated and effective on a real time basis in the applicable areas.

Together we strive to achieve NIL incidents in the entire Hydrocarbon Value Chain. This, besides other issues, calls for total engagement from all levels of the stake holder organizations, which we, at OISD, fervently look forward to.

Jai Hind!!!

Executive Director
Oil Industry Safety Directorate

FOREWORD

The Oil Industry in India is over 100 years old. As such, various practices have been in vogue because of collaboration/ association with different foreign companies and governments. Standardization in design philosophies, operating and maintenance practices remained a grey area. This coupled with feedback from some serious accidents that occurred in the past in India and abroad, emphasized the need for the industry to review the existing state-of-the-art in designing, operating and maintaining of Oil and Gas installations.

With this in view, the Ministry of Petroleum and Natural Gas in 1986 constituted a Safety Council assisted by the Oil Industry Safety Directorate (OISD) staffed from within the industry in formulating and implementing a series of self-regulatory measures aimed at removing obsolescence, standardizing and upgrading the existing standards to ensure safe operations. Accordingly, OISD constituted a number of functional committees of experts nominated from the industry to draw up standards and guidelines on various subjects.

Petroleum Depots, Terminals, Installations; Pipeline Installations; Lube Oil Installations are intermediate processing, storage & transportation points and are vital components of the downstream logistic value chain of the petroleum industry. These play a pivotal role in ensuring seamless movement of product from Refineries to the points-of-sale in the consumption centres. In a vast country like India, with its widespread petroleum product distribution network, continuous safe operation of these units is a national imperative.

For the logistic management purpose, population of these units are large and all of them are located not too far off from human population indicating thereby the high risk perception associated with substantial bulk storage of inflammable petroleum products in such locations. The locations themselves experience heavy footfall of business associates increasing the safety system demands. It is, therefore, essential to develop a standard for fire protection facilities for such locations which would ensure adequacy in terms of its ability to contain any incident from developing into a catastrophe while achieving optimality in terms of life cycle cost of installation, operation and maintenance.

This standard was first released in 1989. Subsequently, with the advent of widespread automation in these types of locations and quantitative increase in storage capacities to satisfy the growth in demand, subsequent revision was issued in Aug 2007. Further amendments to the same were issued in July 2008 and last in October 2010 which emanated out of the in-depth analysis of the recent major fire incident by an expert committee.

This amendment, issued now in July 2012, incorporates the duly optimized recommendations, over and above the October 2010 amendment, emanating out of the in-depth analysis by a series of high power committees constituted in this context in addition to the inputs from the Safety Council of OISD to bring in further clarity after weeding out ambiguities. In this amendment, substantive additions had been made in critical firefighting and mitigating systems after optimizing the same. Attempts had also been made to align the standard to international codes of design, practice and certification.

The amendment issued in October 2017 is related to deletion of educatory text from clause 4.2.7 of the standard.

The figures and photographs used in the document, if any, are representative in nature.

We, at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation of the target group of locations.

Needless to mention, this standard, as always would be reviewed based on field level experience, incident analysis and environment scanning. Suggestions from all stake holders may be forwarded to OISD.

NOTE

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents

Oil & Natural Gas Corporation Limited

FUNCTIONAL COMMITTEE
(Complete Revision: August, 2007)

Name	Organization
Leader	
Shri Arvind Kumar	Engineers India Limited
Members	
1. Shri P. J. Tikekar	Indian Oil Corporation Limited
2. Shri J. Y. Punegar	Hindustan Petroleum Corporation Limited
3. Shri S. P. Garg	GAIL (India) Limited
4. Shri Shashi Dua	Indian Oil Corporation Limited
5. Shri J. Jaisinghani	Indian Oil Blending Limited
6. Shri P. S. Narayanan	Oil India Limited
7. Shri D. K. Banerjee	Indian Oil Corporation Limited
8. Shri S. Ramesh	Bharat Petroleum Corporation Limited
9. Shri A. Rajvanshi	IBP Co. Ltd.
10. Shri M. N. Moharana	Numaligarh Refinery Limited
Co-coordinator	
Shri N. D. Kapaley Up to November, 2002	Oil Industry Safety Directorate
Shri A. Mishra April, 2004 - August, 2007	Oil Industry Safety Directorate
Shri B. R. Gadekar June, 2006 - August, 2007	Oil Industry Safety Directorate

In addition to the above, several other experts from industry contributed in the preparation, review and finalization of this document.

FUNCTIONAL COMMITTEE
(Second Edition: August, 2000)

Name	Organization
Leader	
Shri R. P. Bhatla Up to 31-12-1994	Engineers India Limited
Shri M. M. Kapoor w. e. f. 01-01-1995	Engineers India Limited
Members	
1. Shri R.P. Saxena	Oil & Natural Gas Corporation
2. Shri B. Balan	Hindustan Petroleum Corporation Limited
3. Shri G.S. Wankhede	Bharat Petroleum Corporation Limited
4. Shri C.T. Anantkrishanan	Indian Oil Corporation Limited
5. Shri S. C. Jain	Indian Oil Corporation Limited
6. Shri H.K.B. Singh	IBP Co. Ltd.
7. Shri V. Koti	Indian Oil Corporation Limited (Pipeline)
Co-coordinator	
Shri Vijay M. Ranalkar Up to July, 1996	Oil Industry Safety Directorate
Shri K. S. Ganeshan w. e. f. August, 1996	Oil Industry Safety Directorate

In addition to the above, several other experts from industry contributed in the preparation, review and finalization of this document.

ERRATA AND UPDATION (March 2022)

The Errata-01 dated 11.01.2021 issued for emergency siren code stands withdrawn. It is superseded with 'Errata and Updation' containing the revised guidelines on Emergency Siren Codes based on the PNGRB regulations.

The following contents were updated based on the PNGRB regulations

3.0 DEFINITIONS

3.1.6 EMERGENCY (Added)

Emergency means a situation or scenario which has the potential to cause serious danger to persons, environment, or damage to property and which tends to cause disruption inside or outside the premises and may require the help of outside resources.

CLASSIFICATION OF EMERGENCIES: -

Emergencies can be categorized into three broad levels based on seriousness and response requirements, namely: -

- (a) **Level - I:** This is an emergency or an incident which
 - (i) can be effectively and safely managed, and contained within the site, location, or installation by the available resources;
 - (ii) has no impact outside the site, location, or installation.
- (b) **Level - II:** This is an emergency or an incident which
 - (i) cannot be effectively and safely managed or contained at the location or installation by available resource and additional support is alerted or required;
 - (ii) is having or has the potential to have an effect beyond the site, location, or installation and where external support of mutual aid partner may be involved;
 - (iii) is likely to be a danger to life, the environment, industrial assets, or reputation.
- (c) **Level - III:** This is an emergency or an incident with off-site impact which could be catastrophic and is likely to affect the population, property and environment inside and outside the installation, and management and control is done by district administration. Although the Level-III emergency falls under the purview of District Authority but till they step in, it should be responsibility of the unit to manage the emergency.

Note: Level - I and Level - II shall normally be grouped as an Onsite emergency and Level - III as an Off-site emergency.

5.0 FIRE ALARM/ COMMUNICATION SYSTEM

5.1 FIRE ALARM SYSTEM

- vi) The following emergency siren* codes should be followed for different emergency situations:
 - (a) **Level - I:** The siren code for Level - I shall be decided by an entity depending upon the size and complexity of installation;
 - (b) **Level - II:** A wailing siren for two minutes;

(c) **Level - III:** Same type of siren as in case of Level - II but the same will be sounded for three times at the interval of one minutes i.e. (wailing siren 2 min + gap 1 min + wailing siren 2 min + gap 1 min + wailing siren 2 min) total duration of disaster siren to be eight minutes.

(d) **ALL CLEAR:** Straight run siren for two minutes.

(e) **TEST:** Straight run siren for two minutes at the frequency at least once a week.

* Wherever 'Fire siren' is mentioned in the standard, the same shall be read as 'Emergency siren'.

Oil & Natural Gas Corporation Limited

**FIRE PROTECTION FACILITIES
FOR
PETROLEUM DEPOTS, TERMINALS, PIPELINE INSTALLATIONS
AND
LUBE OIL INSTALLATIONS**

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FIRE PROTECTION FACILITIES FOR PETROLEUM DEPOTS, TERMINALS AND PIPELINE/LUBE OIL INSTALLATIONS

1.0 INTRODUCTION

The petroleum depots, terminals & pipeline/lube oil installations are generally located in the remote areas and near railway sidings. However, the experience shows that with the passage of time, these get surrounded by residential/industrial installations. The inventory of flammable materials stored therein necessitates inbuilt fire protection facilities.

It can be impractical and prohibitively costly to design fire protection facilities to control catastrophic fires. The usual requirement of a good system is to prevent emergencies from developing into major threat to the oil installation and surroundings.

2.0 SCOPE

2.1 This standard lays down the minimum requirement of fire protection facilities at Petroleum Depots, Terminals, Pipeline Installations with or without Storages, Central Tank Farms (CTF). Lube Oil Installations, Grease Manufacturing & Filling Facilities.

2.2 This standard does not cover the fire protection facilities for:-

- i) Depots, Terminals/Installations inside the Refineries and/or Oil/Gas Processing Plants under the same management for which OISD-STD-116 shall be referred.
- ii) Installations handling Liquefied Petroleum Gas (LPG) Storage, Handling and Bottling. The same are covered in OISD-STD-144, OISD-STD-150 or OISD-STD-169 as the case may be.
- iii) Drilling rigs, Work over rigs and Production installations (GGS/OCS, GCP/GCS, EPS, QPS/WHI etc) which are covered in OISD-STD-189.
- iv) Port Oil Terminals for which OISD-STD-156 shall be referred.
- v) Control Room Building & Electrical Installations shall be provided as per OISD-STD-163 & 173 respectively.

2.3 It is intended that the provisions specified in this standard shall be implemented progressively for the existing facilities as given in Annexure-VI of this standard. For the new / upcoming locations/facilities, these shall be implemented along with commissioning. Replacement of the extant Rim Seal fire protection system in the existing Class “A” floating roof tanks would be governed by equivalency clause as given in this standard.

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- 2.4** Wherever specified in the standard, the use of equivalent or superior performance systems / methods to those already prescribed in the standard, should be considered, provided these equivalent systems / methods are duly approved by a technical committee constituted by OISD, based upon technical documentation, performance record and field demonstration by the user industry.

3.0 DEFINITIONS

3.1 PETROLEUM DEPOTS & TERMINALS

A portion of the property, where combustible/flammable liquids are received by tanker, pipelines, tank wagons, tank trucks and are stored or blended in bulk for the purpose of distribution by tankers, pipelines, tank wagons, tank trucks, portable tanks or containers.

3.1.1 PIPELINE INSTALLATIONS

Pipeline Installations are those facilities on cross-country pipelines which have pumping and/or delivery station with or without storages.

3.1.2 LUBE OIL INSTALLATIONS

The facilities meant for receipt, storage and blending of base oils & additives into finished Lube products. It includes lube-blending plants, grease manufacturing plants & small can filling plants.

3.1.3 AVIATION FUELLING STATIONS

The facilities where ATF is received by tank wagons, tank trucks & pipeline and stored in bulk for dispatch of product by refuellers & pipeline. It also includes storage of Methanol/AVGAS & other additives in drums.

3.1.4 INFRASTRUCTURE / OTHER FACILITIES

These are the facilities such as Control Room Building, Sub-Station, Diesel Generator (with diesel storage tank), & Administrative Building, etc. provided in Petroleum Depots, Terminals and Pipeline Installations.

3.1.5 HAZARDOUS AREA

An area will be deemed to be hazardous where:-

Petroleum having flash point below 65°C or any flammable gas or vapor in a concentration capable of ignition is likely to be present.

Petroleum or any flammable liquid having flash point above 65°C is likely to be refined, blended or stored at above its flash point.

For classification and extent of hazardous area, refer "The Petroleum Rules - 2002".

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3.2 CLASSIFICATION OF PETROL-EUM PRODUCTS

3.2.1 Petroleum means any liquid hydrocarbon or mixture of hydrocarbons and any inflammable mixture (liquid, viscous or solid) containing any liquid hydrocarbon.

3.2.2 General Classification

Petroleum products other than LPG which is a separate category are classified according to their closed cup *Flash Point* as follows:-

Petroleum Class A means petroleum having a flash point below 23°C.

Petroleum Class B means petroleum having a flash point of 23°C and above but below 65°C.

Petroleum Class C means petroleum having a flash point of 65°C and above but below 93°C.

Excluded Petroleum means petroleum having a flash point above 93°C and above.

Flash Point of any petroleum means the lowest temperature at which it yields a vapor which will give a momentary flash when ignited.

3.2.3 Classification for Heated Petroleum Products

The locations where product is handled by artificially heating it to above its flash point, Class C product shall be considered as Class B product and Class B product as Class A product.

3.3 GENERAL TERMINOLOGY

Clean agent electrically non-conductive, volatile or gaseous fire extinguishant that does not leave a residue upon evaporation and meets the requirements given in the latest NFPA Standard 2001 (latest edition) on clean agent fire extinguishing systems in line with environmental considerations of Kyoto and Montreal Protocol & latest MoEF regulations.

Shall indicate that provision is mandatory.

Should indicate that provision is recommendatory as per good engineering practices.

GPM denotes US gallons (1GPM=3.785 LPM)

4.0 FIRE PROTECTION PHYLOSOPHY

The fire protection philosophy is based on loss prevention & control. It considers that a depot/terminal carries an inherent potential hazard due to flammable nature of petroleum

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products stored therein. A fire in one facility can endanger other facility of the depot/terminal, if not controlled/extinguished as quickly as possible to minimize the loss of life & property and prevent further spread of fire.

4.1 GENERAL CONSIDERATIONS

The size of product storage & handling facilities, their location and terrain determine the basic fire protection requirements.

4.1.1 Layout

Layout of a depot or terminal, pipeline installation, lube oil installation, grease manufacturing & filling facilities and handling/disposal system of blow down, drain from equipment handling flammable liquids shall be done in accordance with OISD-STD-118 & OISD-STD-109 as applicable.

Special consideration should be given in the plant layout & product line layout for heated products lines laid alongside the pipeline carrying lighter petroleum products.

4.1.2 FIRE PROTECTION

Depending on the nature of risk, following fire protection facilities shall be provided in the installation.

- Fire Water System.
- Foam System.
- Clean Agent Protection System.
- First Aid Fire Fighting Equipment.
- Mobile Fire Fighting Equipment.
- Carbon Dioxide System
- Dry Chemical Extinguishing System
- Portable fire fighting equipment
- Fire Detection, Alarm & Communication System.

4.2 DESIGN CRITERIA FOR FIRE PROTECTION SYSTEM

4.2.1 Facilities shall be designed on the basis that city fire water supply is not available close to the installation.

4.2.2 The fire water system shall be provided based on single largest fire contingency for all locations where total tankage in the terminal is upto 30,000 KL.
The fire water system shall be provided based on two largest fire contingencies simultaneously for all locations where total tankage in the terminal is more than 30,000KL
This clause shall not be applicable for location exclusively storing class C & excluded products.
(Refer Explanatory Note for implementation vide Annexure VI)

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4.2.3 The hazardous areas shall be protected by a well laid combination of hydrants & monitors. The following installations are exempted from this provision:-

- i) The installation having aggregate above ground storage capacity of less than 1000 KL (Class A+B+C) other than AFS.
- ii) Pipeline installation having only scrapper stations or sectionalizing valve stations.

4.2.4 Tank Wagon (TW)/Tank Truck (TT) loading/unloading facilities, Manifold area of product pump house & Exchange pit shall be fully covered with a well laid out combination of hydrants and UL/FM listed/approved variable flow type water-cum-foam monitors.

4.2.5 The installations storing Class A petroleum in above ground tanks shall have fixed water spray system.

However, installations above 1000 KL storage fulfilling the following both conditions are exempted from the provision of fixed water spray system

- Aggregate above ground storage of Class A & B petroleum up to 5000 KL.
- Floating roof tank storing Class A petroleum having diameter up to 9 m.

4.2.6 Class 'B' above ground Petroleum storage tanks (fixed roof or floating roof) of diameter larger than 30 m shall be provided with fixed water spray system.

4.2.7 When Class A & B above ground storage tanks are placed in a common dyke, the fixed water spray system shall be provided on all tanks except for small installations as mentioned in 4.2.5.

4.2.8 TW loading gantries shall be provided with manually operated fixed water spray system. In case automatic fixed water spray system is provided in TW gantry, the gantry shall be divided into suitable number of segments (each segment having min. length of 15 m length & width of 12 m) and three largest segments operating at a time shall be considered as single risk for calculating the water requirement.

Accordingly, a provision shall be made to actuate the water spray system from a safe approachable central location i.e. affected zone and adjoining zones.

4.2.9 The fixed water spray system shall also be provided on all tanks, irrespective of diameter in the installations constructed prior to publication of 1st Edition of OISD-STD-117 in July'1989, where inter distances between tanks in a dyke and/or within dykes are not meeting the requirements of OISD-STD-118.

4.2.10 Fixed foam system or Semi-fixed foam system shall be provided on tanks (floating roof or fixed roof) exceeding 18 m diameter storing Class A or Class B petroleum.

In the installations where inter distances between tanks within a dyke and/or within tank dykes are not conforming to the provisions of OISD-STD-118 and the tanks in the installation are constructed prior to publication of 1st Edition of OISD-STD-117 in July'1989, *fixed foam or semi-fixed foam system shall also be provided on all tanks irrespective of diameter*

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- 4.2.11 Portable foam and/or UL/FM listed/approved variable flow water-cum-foam monitors shall be provided for suppression of pool fire in tank farm area.
- 4.2.12 Automatic actuated Rim Seal fire detection and extinguishing system shall be provided on all existing as well as new external floating roof tank storing Class A Petroleum products.

The rim seal protection system shall be of linear hollow metallic tube type detectors with foam based extinguishing media or equivalent system*, for existing, new installation as well as for replacements of existing system when due. These detection systems shall be listed and/or approved by any of the international agencies like UL, FM, VdS or LPC to ensure that those systems are used which meet with highest standards of safety.

The minimum requirement for design of the system is given in Annexure V.

This is in addition to fixed water spray system and fixed foam system or semi fixed foam system on all floating roof tanks storing class-A & B.

(Refer Explanatory Note for implementation vide **Annexure VI**)

* Refer clause at para 2.4 for equivalency.

- 4.2.13 Fixed water spray system shall also be provided in lube oil drum areas if located in hazardous area.
(Refer Explanatory Note for implementation vide **Annexure VI**)

- 4.2.14 **Clean Agent** (Halon substitute) based flooding system should be provided for control rooms, computer rooms/ repeater station and pressurized rooms in major locations having automated pipeline receipt/dispatch and/or TW/TT loading facilities.

Selection of clean agent and design of fire protection system for control rooms, computer rooms and pressurized rooms should follow the Standard on “Clean Agent Extinguishing systems NFPA Standard 2001 (latest edition) including its safety guidelines with respect to “Hazards to Personnel”, electrical clearance and environmental factors in line with environmental considerations of Kyoto and Montreal Protocol & latest MoEF regulations. (For areas to be covered with clean agent refer OISD STD 163).

- 4.2.15 **Combined POL and LPG facilities in the same premises**

The fire protection and fire fighting requirement for the combined POL and LPG facilities in the same premises shall be based on the following:-

- i. Each POL / LPG facility shall independently meet the design, layout & fire protection system requirements of corresponding OISD standards.
- ii. The fire water requirement shall be based on two fire contingencies simultaneously in the combined facility and fire water storage capacity shall be fixed accordingly.
- iii. The fire water system shall ensure availability of pressure of 7 kg/cm²g at the farthest point.
- iv. The entire fire water system shall remain pressurized and kept in auto mode as recommended for LPG installations.
- v. The control of such facility shall remain with dedicated or LPG group.

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4.3 FIRE WATER SYSTEM

Water is used for fire extinguishments, fire control, cooling of equipment, exposure protection of equipment and personnel from heat radiation.

The fire water ring main shall be provided all around perimeter of the installation with hydrants/monitors spaced at intervals not exceeding 30m when measured aerially.

4.3.1 Components of Fire Water System

The main components of the system are Fire Water Storage, Fire Water Pumps and Distribution Piping Network.

4.3.2 Basis

The fire water system shall be provided based on single largest fire contingency for all locations where total tankage in the terminal is upto 30,000 KL.

The fire water system shall be provided based on two largest fire contingencies simultaneously for all locations where total tankage in the terminal is more than 30,000 KL. This clause shall not be applicable for location exclusively storing class C & excluded products.

(Refer Explanatory Note for implementation vide **Annexure VI**)

4.3.2.1 Design Flow Rate

- i) Fire water flow rate for a tank farm shall be aggregate of the following :-
 - Water flow calculated for cooling a tank on fire at a rate of 3 lpm/m² of tank shell area.
 - Water flow calculated for exposure protection for all other tanks falling within a radius of (R+30) m from centre of the tank on fire (R-Radius of tank on fire) and situated in the same dyke at a rate of 3 lpm/m² of tank shell area.
 - Water flow calculated for exposure protection for all other tanks falling outside a radius of (R+30) m from centre of the tank on fire and situated in the same dyke at a rate of 1 lpm/m² of tank shell area.
 - For water flow calculations, all tanks farms having class A or B petroleum storage shall be considered irrespective of diameter of tanks and whether fixed water spray system is provided or not.
 - Water flow required for applying foam on a single largest tank by way of fixed foam system, where provided, or by use of water/foam monitors. (Refer section 4.4.8 for foam solution application rates.)
 - Various combinations shall be considered in the tank farm for arriving at different fire water flow rate and the largest rate to be considered for design.
- ii) Fire water flow for pump house shed at cross country pipeline installations shall be at a rate of 10.2 lpm/m².

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- iii) Fire water flow rate for TW loading gantry (Refer Annexure – II) and product pump house in a depot or terminal shall be calculated at a rate of @ 10.2 lpm/m².
- iv) For Pump of volatile product/s located under pipe rack fire water flow rate shall be calculated at a rate of 20.4 lpm/m².
- v) Fire water flow rate for supplementary streams shall be based on using 4 single hydrant outlets. Capacity of each hydrant outlet as 36 m³/hr shall be considered at a pressure of 7 kg/cm² (g). The supplementary water stream requirement shall be in addition to design flow rates as calculated in (i), (ii), (iii) & (iv) above.
- vi) The design fire water rate shall be the largest of [4.3.2.1 (i)], [4.3.2.1 (ii)], [4.3.2.1 (iii)], or [4.3.2.1 (iv)]. (Refer Annexure-I).

4.3.3 Header Pressure

Fire water system shall be designed for a minimum residual pressure of 7 kg/cm² (g) at hydraulically remotest point in the installation considering single or double fire contingencies, as the case may be.

4.3.4 Storage

Water for the fire fighting shall be stored in easily accessible surface or underground or above ground tanks of steel, concrete or masonry.

The effective capacity of the reservoir/tank above the level of suction point shall be minimum 4 hours of the aggregate rated capacity of pumps.

For all locations with total storage capacity upto 30,000KL, wherever reliable water replenishment @ 50% or more is available, the fire water storage capacity can be reduced to 3 hours of the aggregate rated capacity of pumps.

For all locations with total storage capacity more than 30,000KL, wherever reliable water replenishment @ 50% or more is available, the single largest fire contingency shall be considered for Fire water storage.

Fresh water should be used for fire fighting purposes. In case sea water or treated effluent water is used for fire fighting purposes, the material of the pipe selected shall be suitable for the service.

The installation shall have facilities for receiving and diverting all the water coming to the installation to fire water storage tanks in case of an emergency.

Storage reservoir shall be in two equal interconnected compartments to facilitate cleaning and repairs. In case of steel tanks there shall be minimum two tanks each having 50 % of required capacity.

Large natural reservoirs having water capacity exceeding 10 times the aggregate fire water requirement can be left unlined.

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4.3.5 Fire Water Pumps

- (i) Fire water pumps having flooded suction shall be installed to meet the design fire water flow rate and head.
If fire water is stored in underground tanks, an overhead water tank of sufficient capacity shall be provided for flooded suction and accounting for leakages in the network, if any.
- (ii) The pumps shall be capable of discharging 150% of its rated discharge at a minimum of 65% of the rated head. The Shut-off head shall not exceed 120% of rated head for horizontal centrifugal pumps and 140% for vertical turbine pump.
- (iii) At least one standby fire water pump shall be provided up to 2 nos. of main pumps. For main pumps 3 nos. and above, minimum 2 nos. standby pumps of the same type, capacity & head as the main pumps shall be provided.
- (iv) The fire water pump(s) including the standby pump(s) shall be of diesel engine driven type. Where electric supply is reliable, 50% of the pumps can be electric driven. The diesel engines shall be quick starting type with the help of push buttons located on or near the pumps or located at a remote location. Each engine shall have an independent fuel tank adequately sized for 6 hours continuous running of the pump.
- (v) Fire water pumps & storage shall be located far away from the potential leak sources / tankage are and shall be at least 30 m (minimum) away from equipment or where hydrocarbons are handled or stored.
- (vi) Fire water pumps shall be exclusively used for fire fighting purpose only.
- (vii) Suction and discharge valves of fire water pumps shall be kept full open all the times.
- (viii) The fire water network shall be kept pressurized by jockey pump(s).
- (ix) A standby jockey pump of similar type capacity and head shall be provided. The capacity of the pump shall be sufficient to maintain system pressure in the event of leakages from valves etc. The capacity of jockey pumps shall be 5% minimum and maximum 10% of the design fire water rate. Auto cut-in / cut-off facility should be provided for jockey pumps.
- (x) The fire water pumps shall be provided with auto cut in facility with pressure drop in fire water network

(Refer Explanatory Note for implementation vide **Annexure VI**)

4.3.6 Fire Water Network

(i) Looping

The fire water network shall be laid in closed loops as far as possible to ensure multi-directional flow in the system. Isolation valves shall be provided in the network to enable isolation of any section of the network without affecting the flow in the rest. The isolation

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valves shall be located normally near the loop junctions. Additional valves shall be provided in the segments where the length of the segment exceeds 300 m.

(ii) Above / Underground Network

The fire water network steel piping should normally be laid above ground at a height of at least 300 mm above finished ground level. Pipes made of composite material shall be laid underground.

However, the ring main shall be laid underground at the following places.

- Road crossings.
- Places where above ground piping is likely to cause obstruction to operation and vehicle movement.
- Places where above ground piping is likely to get damaged mechanically.
- Where frost conditions warrants and ambient temperature is likely to fall subzero, above ground piping shall be laid at least 1 m below the finished grade level to avoid freezing of water. Alternatively, water circulation should be carried out in the above ground pipelines or any other suitable means.

(iii) Protection of underground pipeline

If fire water ring mains are laid underground, the following shall be ensured:-

- The ring main shall have at least 1 m earth cushion in open ground, 1.5 m cushion under the road crossings and in case of crane movement area pipeline shall be protected with concrete/steel encasement as per design requirement.
- For rail crossing, provisions stipulated by Indian Railways shall be complied.
- The under ground ring main shall be protected against soil corrosion by suitable coating/wrapping with or without cathodic protection.
- Pipe supports under the pipe line shall be suitable for the soil conditions.

(iv) Support & Protection of above ground pipelines

The mains shall be supported at regular intervals not exceeding 6 m. For pipeline size less than 150 mm, support interval shall not exceed 3 m.

The pipe support shall have only point contact.

The system for above ground portion shall be analyzed for flexibility against thermal expansion and necessary expansion loops, guides/cross guides and supports provided.

(v) Sizing of pipeline

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- a) The hydraulic analysis of network shall be done at the design time. Also whenever fire water demand increases due to addition of facilities or extensive extension of network, fresh hydraulic analysis shall be carried out.
- b) Fire water ring main shall be sized for 120% of the design water flow rate. Design flow rates shall be distributed at nodal points to give the most realistic way of water requirements in an emergency. It may be necessary to assume several combinations of flow requirement for design of network.

The stand post for hydrants and monitors shall be sized to meet the respective design water flow rates.

(vi) General

Connections for fixed water monitors on the network shall be provided with independent isolation valves.

Fire water mains shall not pass through buildings or dyke areas.

In case of underground mains the isolation valves shall be located in RCC/brick masonry chamber of suitable size to facilitate operation during emergency & maintenance.

4.3.7 Hydrants & Monitors

- i) Hydrants shall be located bearing in mind the fire hazards at different sections of the premises to be protected and to give most effective service. At least one hydrant post shall be provided for every 30 m of external wall measurement or perimeter of battery limit in case of high hazard areas. For non-hazardous area, they shall be spaced at 45 m intervals. The horizontal range & coverage of hydrants with hose connections shall not be considered beyond 45 m.
- ii) Hydrants shall be located at a minimum distance of 15 m from the periphery of storage tank or equipment under protection. In case of buildings this distance shall not be less than 2 m and not more than 15 m from the face of building. Provision of hydrants within the building shall be provided in accordance with IS: 3844.
- iii) Hydrant/Monitors shall be located along road side berms for easy accessibility.
- iv) Double headed hydrants with two separate landing valves or monitor on suitably sized stand post shall be used. All hydrant outlets/monitor isolation valves shall be situated at workable height of 1.2 meter above ground or hydrant/monitor operating platform level.
- v) Monitors shall be located to direct water on the object as well as to provide water shield to firemen approaching a fire.

The requirement of monitors shall be established based on hazards involved and layout considerations. Monitors shall not be installed within 15 m of hazardous equipment.

The location of the monitors shall not exceed 45 m from the hazard to be protected.

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- vi) For marketing terminals & Petroleum Depots, the remote operated high volume long range water cum foam monitors (Capacity 500/750/1000 GPM and above) to fight tank fires shall be provided which shall be of variable flow (with flow adjustable manually in the field). Foam induction to the monitor shall be possible from minimum 60 m distance from the monitor.

For Pipeline & Central crude oil Tank Farms, Manual or / and Remote operated high volume long range water cum foam monitors (capacity 500/750/1000 GPM and above) to fight tank fires shall be provided which shall be of variable flow type (with flow adjustable manually in the field). Foam induction to the monitor shall be possible from minimum 60 m distance from the monitor. Manual type HVLR shall be considered only where sufficient distance is available from the point of hazards and in-company firefighting infrastructure (round the clock fire tender and dedicated manpower) like in Refineries with respect to the location is available.

The location of HVLRs to be planned in such a way that the very purpose of these monitors is served and throw of the monitors is safely delivered at the aimed object. These high volume long range monitors shall be located at distance of 15m to 45 m from the hazardous equipment.

The basic water-cum-foam monitors shall be UL/FM listed/approved. The electrical or hydraulic remote control mechanism shall be in line with Hazardous Area Classification.

Following criteria shall be followed for installation of variable flow foam-cum-water monitors:

- Remote or / and manual operated variable flow monitors shall be installed in such a way that all the tanks in the installation are within the horizontal range of foam throw.
- Minimum two nos. of fixed type HVLR monitors shall be installed for each tank farm containing storage tanks of Class A products having aggregate storage capacity up to 10,000 KL in the installation to meet the requirement as per S.No. (a) above. Monitors shall be placed in opposite direction.
- In tank farm having aggregate storage capacity more than 10,000 KL of Class A product, additional monitor/s should be provided to meet foam application rate of 8.1 LPM/m². Additional monitors shall be provided in such a way that each tank is in the coverage area of at least two monitors.
- For all locations meeting the safety distance norms as per OISD 118, minimum one no. trolley mounted mobile type water cum foam HVLR monitor shall be placed for covering the tank farms storing Class B/C products. In respect of installation storing exclusively Class-C product this provision shall not be applicable.
- For all locations not meeting the safety distance norms as per OISD 118, HVLR monitors of fixed type shall be provided for the tank farms storing Class B/C products also. Requirement of HVLR monitors shall be calculated for full surface fire scenario of the largest tank @8.1 lpm/m². In respect of installation storing exclusively Class-C product this provision shall not be applicable.
- At small installations having aggregate storage capacity upto 10,000 KL, where it is not possible to install the fixed type HVLR monitors at a safe distance (minimum 15 m) from

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the tank because of non availability of adequate space/distance, shall be allowed to provide trolley mounted monitor of suitable capacity for Class A tank also.

- g) Locations where HVLR monitors of fixed type are provided to cover both floating roof and fixed roof tanks, portable type HVLR monitor need not be provided.
- h) Provision for connecting / hooking the portable monitor shall be made in the hydrant system around the fixed roof tanks at various strategic points.
- i) At locations having more than two dykes having storage tanks of Class A products in close proximity, the monitors shall be positioned at suitably convenient location to provide protection to tanks located in different dykes. In such cases, the minimum stipulated requirement of two nos. monitors for each tank farm shall not be required.
- j) Well laid procedures and plans shall be made and put into use for use of HVLRs to combat emergencies without loss of much time.
- k) Numbers & Capacity of monitor shall be provided in such a way that the foam application rate from the monitor meets requirement of foam application rate (8.1 LPM/m²) for full surface tank fire.
- l) For determining the total foam solution requirement, potential foam loss from wind and other factors shall be considered while designing.
- vii) Hydrants and monitors shall not be installed inside the dyke areas. However, as an additional requirement, oscillating monitors shall be provided in inaccessible area within the dyke with isolation valve or ROV outside the tank farm, where inter distances between tanks in a dyke and/or within dykes are not meeting the requirements of OISD-STD-118.
- viii) TW/TT loading & unloading facilities shall be provided with alternate hydrant and UL/FM listed/approved variable flow water-cum-foam monitors having multipurpose combination nozzles for jet, spray & fog arrangement and located at a spacing of 30 m on both sides of the gantry. The hydrants & monitors shall be located at a minimum distance of 15 m from the hazard (e.g. TW & TT loading/unloading facilities) to be protected.
- ix) Hydrants/Monitors shall be located with branch connection.

4.3.8 Material Specifications

The materials used in fire water system shall be of approved type as indicated below:-

i) Pipes

Carbon Steel as per IS: 3589/IS: 1239/IS: 1978 or Composite Material or its equivalent for fresh water service.

In case saline, blackish or treated effluent water is used, the fire water ring main of steel pipes, internally cement mortar lines or glass reinforced epoxy coated or pipes made of material suitable for the quality of water shall be used. Alternately, pipes made of composite materials shall be used.

The composite material to be used shall be as per API 15LR/API 15HR.

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ii) Isolation Valves

Gate or butterfly type isolation valves made of Cast Steel having open/close indication shall be used. Other materials such as cupro-nickel for saline/blackish water can be used.

iii) Hydrants

- Stand post - Carbon Steel
- Outlet valves - Gunmetal/ Aluminum/ Stainless/ Steel/Al-Zn Alloy

iv) Monitors

As per UL/FM listed/approved.

v) Fire Hoses

Reinforced Rubber Lined Hose as per IS 636 (Type A)/Non-percolating Synthetic Hose (Type B)/UL or Equivalent Standard.

(vi) Fire water mains, hydrant & monitor stand posts, risers of water spray system shall be painted with "Fire Red" paint as per of IS: 5.

vii) Hose boxes, water monitors and hydrant outlets shall be painted with "Luminous Yellow" paint as per IS: 5.

viii) Corrosion resistant paint shall be used in corrosion prone areas.

4.3.9 FIXED WATER SPRAY SYSTEM

- i) Fixed water spray system is a fixed pipe system connected to a reliable source of water supply and equipped with water spray nozzles for specific water discharge and distribution over the surface of area to be protected. The piping system is connected to the hydrant system water supply through an automatically or manually actuated valve which initiates the flow of water. In case the system is manually actuated, the isolation valve shall be located outside the dyke for ease of access & operation.
- ii) Spray nozzles shall be directed radially to the tank at a distance not exceeding 0.6 m from the tank surface. Only one type and size of spray nozzle shall be used in a particular facility.
- iii) While calculating the water rates for spray application for cases other than tanks/vessels, the area should be divided into suitable segments so that maximum water requirement can be optimized. (Refer Annexure-II for typical calculations).

4.4 FOAM SYSTEMS

4.4.1 Types of Foam

Foams are classified by producing action of generation and expansion. Foam concentrate to be used shall conform to IS: 4989 2006/UL-162 or Equivalent Standard (Annexure – IV)

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4.4.2 Types of Low Expansion Foam

For combating large hydrocarbon fires particularly in a contained area like storage tank, foam has proved useful for its inherent blanketing ability, heat resistance and security against burn-back. Aqueous Film Forming Foam (AFFF) compound is technically superior and compatible with other fire fighting agents.

Efficient and effective foam delivery system is a vital tool for its usefulness in controlling the fire.

The process of adding or injecting the foam concentrate to water is called proportioning. The mixture of water and foam compound (foam solution) is then mixed with air in a foam maker for onward transmission to burning surface.

4.4.3 CONVEYING SYSTEMS

The system consists of an adequate water supply, supply of foam concentrate, suitable proportioning equipment, a proper piping system, foam makers and discharge devices designed to adequately distribute the foam over the hazard.

Conventional systems are of the open outlet type, in which foam discharges from all foam outlets at the same time, covering the entire hazard within the confines of the system.

There are three types of systems:-

- i) Fixed
- ii) Semi-Fixed
- iii) Mobile

(i) Fixed Foam System

Fixed foam conveying system comprises of fixed piping for water supply at adequate pressure, foam concentrate tank, eductor, suitable proportioning equipment for drawing foam concentrate and making foam solution, fixed piping system for onward conveying to foam makers for making foam, vapor seal box and foam pourer.

(ii) Semi-Fixed Foam System

Semi-fixed foam system gets supply of foam solution through the mobile foam tender. A fixed piping system connected to foam makers cum vapor seal box in case of cone roof tanks and foam maker and foam pourers in the case of floating roof tanks conveys foam to the surface of tank.

(iii) Mobile System

Mobile system includes foam producing unit mounted on wheels which should be self propelled or towed by a vehicle. These units supply foam through monitors/foam towers to the burning surface.

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(iv) Sub-surface foam injection

This system is for protection of fixed roof storage tanks. It comprises of high back pressure foam generator connected through product lines or separate lines near the bottom of the tank.

(v) Under the Seal Foam application

This is a system for floating roof tank where the foam travels through a flexible pipe inside the tank up to the center of the tank roof and exits at the seal rim of the floating roof precisely where the fire is located thus rapidly flooding the seal rim area and quickly extinguishing the fire.

4.4.4 FLOATING ROOF TANK PROTECTION

For floating roof tank, foam shall be poured at the foam dam to blanket the roof seal. Features of foam system for floating roof tank protection shall be as follows:-

- i) System shall be designed to create foam blanket on the burning surface in a reasonably short period.
- ii) Foam shall be applied to the burning hazard continuously at a rate high enough to overcome the destructive effects of radiant heat.
- iii) Foam makers/foam pourers shall be located not more than 24 M apart on the shell perimeter based on 600 mm foam dam height. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal.
- iv) A minimum of two foam pourers shall be provided.

Automatic Actuated Rim seal Protection System for Floating Roof tanks:

Automatic actuated Rim Seal fire detection and extinguishing system shall be provided on all existing as well as new external floating roof tank storing Class A Petroleum products.

The rim seal protection system shall be of linear hollow metallic tube type detectors with foam based extinguishing media or equivalent system*, for existing, new installation as well as for replacements of existing system when due. These detection systems shall be listed and/or approved by any of the international agencies like UL, FM, VdS or LPC to ensure that those systems are used which meet with highest standards of safety.

The minimum requirement for design of the system is given in Annexure V.

This is in addition to fixed water spray system and fixed foam system or semi fixed foam system on all floating roof tanks storing class-A & B.

(Refer Explanatory Note for implementation vide **Annexure VI**)

* Refer clause at para 2.4 for equivalency.

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4.4.5 FIXED ROOF TANK PROTECTION

Foam conveying system shall have same features as of floating roof tank excepting that a vapor seal chamber is required before the foam discharge outlet.

Features of the foam system for fixed roof protection shall be as follows:

- i) The vapor seal chamber shall be provided with an effective and durable seal, fragile under low pressure, to prevent entrance of vapor into the foam conveying piping system.
- ii) Where two or more pourers are required these shall be equally spaced at the periphery of the tank and each discharge outlet shall be sized to deliver foam at approximately the same rate.
- iii) Tanks should be provided with foam discharge outlets/pourers as indicated below :-

Tank diameter (In M)	Foam Pourer (Min. Nos.)
Above 18 & up to 20	2
Above 20 & up to 25	3
Above 25 & up to 30	4
Above 30 & up to 35	5
Above 35 & up to 40	6
Above 40 & up to 45	8
Above 45 & up to 50	10

In case foam pourers are provided on tanks having diameter up to 18 m, minimum 2 nos. foam pourers shall be provided.

The estimation of number of foam discharge outlet is based on pourer capacity of 1000 lpm at a pressure of 7 kg/cm² (g) upstream of eductor. This can be suitably adjusted for different pourer capacity in accordance with section 4.4.4 (iii).

4.4.6 FLOATING CUM FIXED ROOF TANK PROTECTION

Protection facilities shall be provided as required for fixed roof tank.

4.4.7 PROTECTION FOR DYKE AREA/SPILL FIRE

Portable monitors/foam hose streams shall be provided for fighting fires in dyked area and spills. Additionally, Medium expansion foam generators shall be provided to arrest vapor cloud formation from spilled volatile hydrocarbons.

Installation of medium expansion foam generator shall be as per following criteria:

Class A tanks:

2 nos. Fixed type foam generators for each tank dyke.

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Class B tanks:

Two nos. portable foam generator for each location.

(Refer Explanatory Note for implementation vide **Annexure VI**)

4.4.8 FOAM APPLICATION RATE

The minimum delivery rate for primary protection based on the assumption that all the foam reaches the area being protected shall be as indicated below :-

For cone roof tanks containing liquid hydrocarbons, the foam solution delivery rate shall be at least 5 lpm/ m² of liquid surface area of the tank to be protected.

For floating roof tanks containing liquid hydrocarbons foam solution delivery rate shall be at least 12 lpm/ m² of seal area with foam dam height of 600 mm of the tank to be protected. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal. In the case of Floating roof tank roof sinking, the application rate shall be considered as 8.1 lpm/ m².

In determining total solution flow requirements, potential foam losses from wind and other factors shall be considered.

4.4.9 DURATION OF FOAM DISCHARGE

The equipment shall be capable of providing primary protection at the specified delivery rates for the following minimum duration.

- i) Tanks containing Class 'A' & 'B' - 65 minutes.
- ii) Where the system's primary purpose is for spill fire protection - 30 minutes.

4.4.10 WATER FOR FOAM MAKING

Water quantity required for making foam solution depends on the percent concentration of foam compound. Foams in normal use have a 3% to 6% proportioning ratio. However, foam supplier data shall be used for determining water requirement.

4.4.11 FOAM QUANTITY REQUIREMENT

The aggregate quantity of foam solution should be calculated as below:-

- i) Foam solution application at the rate of 5 lpm/ m² for the liquid surface of the single largest cone roof tank or at the rate of 12 lpm/ m² of seal area of the single largest floating roof tank whichever is higher. (Annexure-III).
- ii) Based on the size of the terminal, quantity of foam solution required should be calculated as per the following guidelines:-

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Size of Terminal (In KL)	Water/Foam Monitor (Number)
For installation having Aggregate capacity of 1000KL	Nil.
For Installation having aggregate capacity above 1000KL & upto 10,000 KL	Minimum 2 Numbers as per 4.3.7(vi)
For Installation having aggregate capacity more than 10,000 KL	More than 2 Numbers as per 4.3.7(vi)c

- iii) Two hose streams of foam each with a capacity of 1140 lpm of foam solution.

The aggregate quantity of foam solutions should be largest of 4.4.11(i), 4.4.11(ii) and 4.4.11 (iii) as above for a minimum period of 65 minutes. From this the quantity of foam based on 3% or 6% proportion should be calculated.

However, for installation having aggregate storage not more than 10,000 KL, the foam concentrate storage shall be based on 4.4.1(i) only.

In case of Aviation Fuelling Stations where aggregate product storage capacity is less than 1000 KL, foam quantity for spill fire protection of 30 minutes shall be made.

Additional Foam quantity requirement & foam monitor requirement shall be in line with 4.3.7(vi) d.

4.4.12 FOAM COMPOUND STORAGE

Foam compound should be stored as explained in IS-4989:2006/UL-162.

Type of foam compound to be used can be protein, fluoro-protein or AFFF. Alcohol Resistant Foam shall be used for handling methanol/ ethanol or furfural fires. Minimum 1000 liter of Alcohol Resistant Foam compound shall be maintained at the installation to handle methanol/ethanol or furfural fire.

Shelf life of foam compound shall be taken from manufacturer's data.

Foam compound shall be tested periodically as per OEM guidelines to ensure its quality and the deteriorated quantity replaced. The deteriorated foam compound can be used for fire training purposes. For details of type of tests & their periodicity, refer IS 4989: 2006/UL-162 or Equivalent Standard.

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Quantity of foam compound equal to 100% of requirement as calculated in 4.4.11 should be stored in the Installation. This quantity can be suitably reduced, if mutual aid for foam supply is available. For sample calculation, refer (Annexure- III).

4.5 CONTROL ROOM AND COMPUTER ROOM PROTECTION

Control room and computer room should be protected by Clean Agent Fire Extinguishing System.

It is considered good practice to avoid unnecessary exposure to Clean Agent Fire Extinguishing System. In order to minimize the exposure, persons should be evacuated from the areas before the system comes into operation.

Clean agent fire extinguishing system as per NFPA Standard 2001 (latest edition) shall be provided for such protection system.

Each hazard area to be protected by the protection system shall have an independent system.

The time needed to obtain the gas for replacement to restore the systems shall be considered as a governing factor in determining the reserve supply needed. 100% standby containers shall be considered for each protected hazard.

Storage containers shall be located as near as possible to hazard area but shall not be exposed to fire.

Storage containers shall be carefully located so that they are not subjected to mechanical, chemical or other damage. All the components of the system shall be capable of withstanding heat of fire and severe weather conditions.

4.6 FIRST AID FIRE FIGHTING EQUIPMENT

4.6.1 Portable Fire Extinguishers

- i) All fire extinguishers shall conform to respective IS/UL or Equivalent codes, viz. 10 Kg DCP Type (IS: 15683 /UL 299), 4.5/6, 8 Kg CO₂ Type (IS: 2878/UL 154) & 25/50/75 Kg DCP Type (IS: 10658/UL 299) and bear ISI/UL mark. BIS/UL or Equivalent certificates of all extinguishers shall be maintained at the location.
- ii) While selecting the Extinguisher, due consideration should be given to the factors like flow rate, discharge time and throw in line with IS: 2190 / UL 711.
- iii) The Dry Chemical Powder used in extinguisher and carbon dioxide gas used as expelling agent shall be as per relevant IS/UL or Equivalent code.
- iv) While selecting the dry chemical powder, due consideration should be given to the typical properties viz. Apparent Density (0.65 +/- 0.05), Fire Rating (144B), Thermal Gravimetric Analysis (with decomposition at around 250°C) and foam compatibility.

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- v) Siliconised Potassium bicarbonate DCP powder (IS 4308:2003) / Mono-ammonium phosphate based DCP powder (IS: 14609) can also be used for recharging DCP fire extinguishers.
- vi) Spare CO₂ cartridges and DCP refills as required based on their shelf life should be maintained. However, minimum 10% of the total charge in the extinguishers should be maintained at the location.
- vii) Portable fire extinguishers shall be located at convenient locations and are readily accessible and clearly visible at all times.
- viii) The sand buckets shall have round bottom with bottom handle
having 9 liter water capacity conforming to IS: 2546. The sand stored in bucket shall be fine and free from oil, water or rubbish.
- ix) Rain protection of suitable design should be provided for all extinguishers & sand buckets.
- x) The maximum running distance to locate an extinguisher shall not exceed 15 m.
- xi) The extinguisher shall be installed in such a way that its top surface is not more than 1.5m above the floor/ground level.
- xii) The no. of extinguishers at various locations shall be provided as under.

Petroleum Depots, Terminals & Lube Oil Installations

Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(i)	Lube Godown	1 No. 10 Kg DCP extinguisher for every 200 m ² or min. 2 Nos. in each Godown whichever is higher.
(ii)	Lube Filling Shed	1 No. 10 Kg DCP extinguisher for 200 m ² or min. 2 Nos. in each Shed whichever is higher
(iii)	Storage of (Class A/B) in packed containers and stored in open/closed area.	1 No. 10 Kg DCP extinguisher for 100 m ² or min. 2 Nos. in each Storage Area whichever is higher.
(iv)	Pump House (Class A/B) Up to 50 HP Above 50-100 HP Beyond 100 HP	1 No. 10 Kg DCP for 2 pumps. 1 No. 10 Kg DCP for each pump. 2 Nos. of 10 kg or 1 no. of 25 kg DCP for each pump.
(v)	Pump House (Class C) Up to 50 HP Above 50 HP	1 no. 10Kg DCP for every 4 pumps up to 50 HP. 2 nos. 10 Kg DCP or 1x25 kg DCP for 4 pumps.
(vi)	Tank Truck loading & unloading gantry for POL/Special products	1 No. 10 Kg DCP extinguisher for each bay plus 1 No. 75 Kg DCP extinguisher for each gantry.
(vii)	Tank Wagon loading and unloading gantry/siding	1 No. 10 Kg DCP extinguisher for every 30 m of gantry/siding plus 1 No. 75 Kg DCP extinguisher for each gantry/siding.
(viii)	A/G Tank Farm	2 Nos. 10 Kg DCP extinguishers for each tank plus 4 Nos. 25 Kg DCP extinguishers for each Tank Farm positioned at four corners. In case of adjoining tank farms, the no. of 25 Kg extinguishers can be reduced by 2 nos. per tank farm.

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Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(ix)	U/G Tank Farm	2 Nos. 10 Kg DCP extinguisher for each Tank Farm
(x)	Other Pump Houses	1 No. 10 Kg DCP extinguisher for every two pumps or min 2 Nos. 10 Kg DCP extinguisher for each Pump House whichever is higher.
(xi)	Admin. Building/Store House	1 No. 10 Kg DCP extinguisher for every 200 m ² or min. 2 Nos. 10 Kg DCP extinguishers for each floor of Building/Store whichever is higher.
(xii)	DG Room	2 Nos. each 10 Kg DCP & 4.5 Kg CO ₂ extinguishers for each DG room.
(xiii)	Main switch Room/Sub-Station	1 No. 4.5 Kg CO ₂ extinguisher for every 25 m ² plus 1 No. 9 Liter sand bucket.
(xiv)	Computer Room/ Cabin	2 Nos. of 2 Kg CO ₂ or 2 Nos. of 2.5 Kg Clean Agent extinguisher per Computer Room and 1 No. 2 Kg CO ₂ or 1 No. 1.0 Kg Clean Agent extinguisher per cabin.
(xv)	Security Cabin	1 No. 10 Kg DCP extinguisher per cabin.
(xvi)	Canteen	1 No. 10 Kg DCP extinguisher for 100 m ² .
(xvii)	Workshop	1 No. 10 Kg DCP extinguisher & 1 No. 2 Kg CO ₂ extinguisher.
(xviii)	Laboratory	1 No. 10 Kg DCP extinguisher & 1 No. 4.5 Kg CO ₂ extinguisher.
(xix)	Oil Sample Storage Room	1 No. 10 Kg DCP extinguisher per 100 m ² or min. 1 no. 10 Kg extinguisher per room whichever is higher.
(xx)	Effluent Treatment Plant	1 No. 75 Kg. & 2 nos. 10 Kg. DCP Extinguisher
(xxi)	Transformer	1 No. 10 Kg. DCP extinguisher per transformer.
(xxii)	UPS / Charger Room	1 No. 2 Kg. CO ₂ extinguisher.

NOTE: - ALL FIRE EXTINGUISHERS SHALL BEAR ISI OR EQUIVALENT MARK

Pipeline Installations

For pipeline installations, the portable extinguisher shall be provided as per the above list (4.6.1) suitably amended along with following additions:-

Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(i)	Main line pump shed (Engine/Motor Driven)	1 No. 75 Kg DCP, 10 Kg DCP & 6.8 Kg CO ₂ extinguishers per two pumps up to a maximum of 4 nos.
(ii)	Booster Pump	1 No. 10 Kg DCP per two pumps up to a maximum of 3 nos. and 1 No. 6.8 Kg CO ₂ extinguisher.
(iii)	Sump Pump, Transmix Pump & Oil Water Separator Pump	1 No. 10 Kg DCP extinguisher.
(iv)	Scraper Barrel	1 No. 10 Kg DCP extinguisher.
(v)	Control Room	2 Nos. 2.5 Kg Clean Agent and 1 No. 4.5 Kg CO ₂ extinguisher.
(vi)	UHF / Radio Room	2 Nos. 2.5 Kg Clean Agent and 1 No. 4.5 Kg CO ₂ extinguisher.

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(vii)	Meter Prover/Separator Filter	1 No. 10 Kg DCP extinguisher.
(viii)	Repeater Station	1 No. 10 Kg DCP & 1 No. 2 Kg CO2 extinguisher.
(ix)	Mainline Emergency Equipment Centre	4 Nos. 10 Kg DCP & 2 Nos. 2 Kg CO2 extinguishers.
(x)	Air Compressor	1 No. 2 Kg CO2 & 1 No. 5 Kg DCP extinguisher.

4.6.2 Wheeled Fire Fighting Equipment

For Installations having tanks of diameter larger than 9 m, following fire fighting equipment shall be provided:-

Size of Terminal (In KL)	Water/Foam Monitor (Nos.)
For installation having aggregate capacity of 1000 KL	Nil.
For installation having aggregate capacity up to 10,000 KL	Minimum 2 Number of 1000 GPM each
For installation having aggregate capacity more than 10,000 KL	More than 2 Numbers of 1000 GPM each

Foam compound trolley 200/210 liters shall be provided as under:-

Tank diameter (In m)	Foam compound trolley (Nos.)
Up to 24 m	1 No.
24 m - 30 m	2 Nos.
Above 30 m	3 Nos.

4.6.3 HOSES, NOZZLES & ACCESSORIES

(i) Hoses

- i) Reinforced rubber lined canvas or Non-percolating synthetic fire hoses conforming to IS- 636/UL 19 (Type A or B) shall be provided.
- ii) The length and diameter of the hoses shall be 15 m and 63 mm respectively fitted with instantaneous type male & female couplings of material as specified in IS 636/UL 19.
- iii) The number of hoses stored in an oil installation shall be 30% of the number of hydrant outlets. The minimum No. of hoses stored, however, shall not be less than 10.
- iv) The hoses shall be stored at convenient and easily accessible location in the oil installation.

(ii) Nozzles

In addition to the jet nozzle provided in each hose box, there shall be at least two nozzles in each category viz. Jet nozzle with branch pipe, Fog nozzle, Universal nozzle, Foam branch pipe and Water curtain nozzle as per relevant IS/UL Codes maintained at the location.

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(iii) Accessories

The following minimum no. of Personal Protective Equipment, First Aid Equipment & Safety Instrument shall be provided as indicated against each item.

Sand drum with scoop: 4 Nos.
 Safety helmet: 1 No. per person.
 Stretcher with blanket: 2 Nos.
 First Aid box: 1 No.
 Rubber hand glove: 2 Pairs.
 Explosimeter : 1 No.
 Fire proximity suit: 1 Suit.
 Resuscitator: 1 No.
 Electrical siren (3 Km range): 1 No.
 Hand operated siren: One each at strategic locations such as Admn Bldg, Laboratory, T/L Loading/Unloading Facility, T/W Loading/Unloading Facility, Tank Farm, FW Pump House & Product Pump House (s).
 Water jel blanket: 1 No.
 Red & Green flag for fire drill: 2 Nos. in each color.
 SCBA Set (30 minute capacity): 1 set with spare cylinder.
 PA system - 1 No.
 Hose box: Between two hydrant points.
 Fire hose: 2 Nos. per hose box.
 Jet nozzle: 1 No. in each hose box.

The above guidelines are minimum requirement of each item and can be increased depending on the scale of operations/size of installation or requirement of Local Statutory Bodies/State Govt.

A trolley containing Fire Proximity Suit, B. A. Set, Water Jel Blanket, Resuscitator, First Aid Box, Stretcher with blanket, Spare fire hoses, Special purpose nozzles, Foam branch pipes, Explosive meter, P. A. System shall be readily available at the location and positioned to have easy access to it during emergency situation.

In addition, an emergency kit shall be provided consisting of safety items as per the attached list (**Annexure VII**) and shall be readily available at the terminals.

All the items of the kit shall be kept on a trolley specifically designed for the purpose. List of PPE's as mentioned in other clauses should be merged with this list.

(Refer Explanatory Note for implementation vide **Annexure VI**)

4.7 MOBILE FIRE FIGHTING EQUIPMENT

Mobile fire fighting equipments include Foam trolleys, Portable water-cum-foam monitors, etc. In view of comprehensive Fixed and First Aid Fire protection equipment recommended in the

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standard, provision of Mobile fire fighting equipments in the installation is not considered necessary. However, the requirement of such equipment should be reviewed keeping in mind the size, nature and location of the installation.

5.0 FIRE ALARM/COMMUNICATION SYSTEM

5.1 FIRE ALARM SYSTEM

- i) Hand operated sirens shall be provided at strategic locations and clearly marked in the installation.
- ii) Electric fire siren shall be installed at suitable location with operating switch located near the risk area at a safe, identifiable and easily accessible place.
- iii) Electric fire siren shall be audible to the farthest distance in the installation and also in the surrounding area up to 1 km from the periphery of the installation.
- iv) Electric fire sirens shall be connected to feeder to ensure continuous power supply during emergency shut down.
- v) The tone of fire siren shall be different from shift siren.
- vi) The following fire siren codes should be followed for different emergency situations.
 - **FIRE:** For fire situation, the siren shall be wailing sound for 2 minutes.
 - **DISASTER:** For disaster situation, the siren shall be wailing sound for 2 minutes repeated thrice with a gap of 10seconds.
 - **ALL CLEAR:** For all clear situation, the siren shall be straight run sound for 2 minutes.
 - **TEST SIREN:** For testing, the siren shall be straight run sound for 2 minutes.

5.2 COMMUNICATION SYSTEM

- i) Communication system like Telephone, Public Address System, etc. should be provided in non-hazardous areas of the installation.
- ii) In hazardous areas, flame-proof/intrinsically safe Paging System, Walkie-talkie system or VHF Set shall be provided.
- iii) Wherever possible hot line connection between City Fire Brigade & nearby industries shall be provided for major installation on need basis.

5.3 DETECTION AND ALARM SYSTEM

Hydrocarbon detectors shall be installed near all potential leak source of class-A e.g. tank dykes, tank manifolds, pump house manifold etc. (Refer Explanatory Note for implementation vide **Annexure VI**)

Hydrocarbon detector of proper type shall be selected and also shall be proof tested and shall be maintained in good condition.

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6.0 FIRE SAFETY ORGANISATION/ TRAINING

6.1 ORGANISATION

A well defined comprehensive On-site Emergency Plan as per OISD-GDN-168 shall be drawn.

6.2 TRAINING

- i) The safety, rescue operation and fire fighting training shall be compulsory for all officers, operators, security, T/T drivers & contract workmen, clericals who are likely to be present in the installation & record maintained. The training shall be conducted through oil industry approved reputed institute (Refer Explanatory Note for implementation vide **Annexure VI**)
- ii) Every employee or authorized person of contractor working in the installation shall be familiarized with fire siren codes and the location of fire siren operating switch nearest to his place of work.
- iii) Instructions on the action to be taken in the event of fire should be pasted at each siren point and familiarity with these instructions ensured and recorded.
- iv) Monthly fire drills considering various scenarios shall be conducted regularly with full involvement of all employees of the installation. The mock drill shall include the full shut down system activation once in six months. (Refer Explanatory Note for implementation vide **Annexure VI**)
- v) The offsite disaster mock drills shall be conducted periodically as per local statutory requirements. The company should approach and coordinate with the district authority for conducting "Offsite Mock Drills".
- vi) The post drill analysis should be carried out & discussed emphasizing areas of improvements.
- vii) The record of such drills should be maintained at the location.
- viii) Security staff should be trained as first responders for fire fighting and rescue operation along with plant operating personnel through oil industry approved reputed institute.

6.3 Mutual Aid:

Installation shall have a 'Mutual Aid' arrangement with nearby industries to pool in their resources during emergency.

7.0 FIRE EMERGENCY MANUAL

- i) Each installation shall prepare a Comprehensive fire emergency manual covering all emergency scenarios outlining the actions to be taken by each personnel in the event of fire emergency for effective handling and the same shall be available to all personnel in the installation.
- ii) The key action points of this manual shall be displayed at strategic locations in the installation for ready reference.

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8.0 FIRE PROTECTION SYSTEM, INSPECTION AND TESTING

- i) The fire protection equipment shall be kept in good working condition all the time.
- ii) The fire protection system shall be periodically tested for proper functioning and logged for record and corrective actions.
- iii) One officer shall be designated and made responsible for inspection, maintenance & testing of fire protection system.
- iv) The responsibilities of each officer shall be clearly defined, explained and communicated to all concerned in writing for role clarity.
- v) In addition to the following routine checks/maintenance, the requirements of OISD-STD-142 in respect of periodic inspection, maintenance & testing of fire fighting equipment shall be complied with.

8.1 FIRE WATER PUMPS

- i) Every pump shall be test run for at least half an hour or as per OEM guidelines, whichever is higher twice a week at the rated head & flow.
- ii) Each pump shall be checked, tested and its shut-off pressure observed once in a month.
- iii) Each pump shall be checked & tested for its performance once in six month by opening required nos. of hydrants/monitors depending on the capacity of the pump to verify that the discharge pressure, flow & motor load are in conformity with the design parameters.
- iv) Each pump shall be test run continuously for 4 hours at its rated head & flow using circulation line of fire water storage tanks and observations logged once a year.
- v) The testing of standby jockey pump, if provided shall be checked weekly. Frequent starts & stops of the pump indicate that there are water leaks in the system which should be attended to promptly.

8.2 FIRE WATER RING MAINS

- (i) The ring main shall be checked for leaks once in a year by operating one or more pumps & keeping the hydrant points closed to get the maximum pressure.
- (ii) The ring mains, hydrant, monitor & water spray header valves shall be visually inspected for any missing accessories, defects, damage and corrosion every month and records maintained.
- (iii) All valves on the ring mains, hydrants, monitors & water spray headers shall be checked for leaks, smooth operation and lubricated once in a month.

8.3 FIRE WATER SPRAY SYSTEM

- i) Water spray system shall be tested for performance i.e. its effectiveness & coverage once in six months.

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- ii) Spray nozzles shall be inspected for proper orientation, corrosion and cleaned, if necessary at least once a year.
- iii) The strainers provided in the water spray system shall be cleaned once in a quarter and records maintained.

8.4 FIXED/SEMI FIXED FOAM SYSTEM

Fixed/Semi fixed foam system on storage tanks should be tested once in six months. This shall include the testing of foam maker/chamber.

The foam maker/chamber should be designed suitably to facilitate discharge of foam outside the cone roof tank. After testing foam system, piping should be flushed with water.

8.5 CLEAN AGENT SYSTEM

Clean agent fire extinguishing system should be checked as under:-

- i) Agent quantity and pressure of refillable containers shall be checked once every **six** month.
- ii) The complete system should be inspected for proper operation once every year (Refer NFPA Standard 2001 - latest edition) for details of inspection of various systems.

8.6 HOSES

Fire hoses shall be hydraulically tested once in six months to a water pressure as specified in relevant IS/UL/Equivalent codes.

8.7 COMMUNICATION SYSTEM

Electric and hand operated fire sirens should be tested for their maximum audible range once a week.

8.8 FIRE WATER TANK/RESERVOIR

- i) Above ground fire water tanks should be inspected externally & internally as per OISD-STD-129.
- ii) The water reservoir shall be emptied out & cleaned once in 3 years. However, floating leaves, material or algae, if any shall be removed once in 6 months or as & when required.

8.9 FIRE EXTINGUISHERS

Inspection, testing frequency and procedure should be in line with OISD-STD-142.

9.0 COMMON FIRE FIGHTING FACILITY FOR CLUSTER OF TERMINALS

Where there is cluster of POL terminals of different companies, provision of jointly owned common fire fighting facility may be considered. The cluster of terminals should be treated

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as single entity for the purpose of designing fire fighting facility, considering one of the following two categories as the case may be.

(a) Total tankage upto 30,000 KI.

(b) Total tankage more than 30,000 KI.

The common fire fighting facility shall be located at a safe distance, beyond the blast overpressure zone.

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10.0 REFERENCES

- 1) NFPA 11 - Standard on Low, Medium and high Expansion Foam Systems
- 2) NFPA 13 - Standard on Installation of Sprinkler System.
- 3) NFPA 15 - Standard on Installation of Water Spray System.
- 4) NFPA 20 - Standard on Installation of Centrifugal Fire Pumps.
- 5) NFPA 2001 - Standard on Clean Agent Fire Extinguishing System.
- 6) No.72-289 -French Regulation for Hydrocarbon Depots.
- 10) The Petroleum Rules –2002.
- 11) Model Code of Safe Practices The Institute of Petroleum (U.K.)
- 15) International Safe Practices of Oil Industry.
- 16) IS-3844: Code of Practice on Installation of Internal Hydrants in Multistory Building.
- 17) OISD-GDN-115: Guidelines on Fire Fighting Equipment & Appliances in Petroleum Industry.
- 18) OISD-STD-142: Standard on Inspection of Fire Fighting Equipment & Systems.
- 19) OISD-STD-154: Standard on Safety Aspects in Functional Training.
- 20) Ozone Depletion Substances Regulation & Control Rules - 2000 Ministry of Environment & Forests, Government of India.
- 21) Kyoto & Montreal Protocols.
- 22) IS-15683: Standard on Portable Fire Extinguishers - Performance & Construction - Specifications.
- 23) IS: 4308: Standard on Dry Chemical Powder for Fighting B & C Class Fires – Specifications.
- 24) IS: 14609: Standard on Dry Chemical Powder for Fighting A, B, C Class Fires – Specifications.
- 25) IS: 4989: Standard on Foam Concentrate for Producing Mechanical Foam for Fire-Fighting Flammable Liquid Fires - Specifications

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ANNEXURE- I

SAMPLE CALCULATION OF FIRE WATER FLOW RATE ^{Note1}
(Clause 4.3.2 to be applicable)

1. DESIGN BASIS

The fire water system in an installation shall be designed to meet:

- i) The fire water flow requirement of fighting single largest fire contingencies for locations where total aggregated storage capacity in the location is upto 30,000KL
- ii) The fire water flow requirement of fighting two largest fire contingencies simultaneously for all locations where total aggregated storage capacity in the terminal is more than 30,000 KL.

2. FIRE WATER DEMAND FOR SINGLE LARGEST FIRE
(For locations with aggregate storage capacity upto 30000KL)

Consider various areas under fire and calculate fire water demand for each area based on design basis as indicated below, however, actual tank dimensions available in the terminal shall be considered.

2.1 FIRE WATER FLOW RATE FOR FLOATING ROOF TANK PROTECTION**Data**

Total storage capacity in one dyke area	=	20,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	10,000 m ³ .
Diameter of each tank	=	30 m.
Height of each tank	=	14.4 m.

a) Cooling water flow rate**(i) Cooling water required for tank on fire**

Cooling water rate	=	3 lpm/m ² of tank area for tank on fire.
Cooling water required	=	$\pi \times 30 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2$.
	=	4073.1 lpm.
	=	$\frac{4073.1 \times 60}{1000} \text{ m}^3/\text{hr} = 244 \text{ m}^3/\text{hr}.$

Assuming that second tank is also located within the same tank dyke at a distance more than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 1 lpm/m² of tank shell area.

Note-1: These are sample calculations only. Calculations on the basis of actual site conditions and dimensions need to be carried out for each installation as per guidelines provided in clause 4.3.

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(ii) **Cooling water required for tank falling beyond (R+30) from centre of tank on fire**

$$\begin{aligned}
 \text{Cooling water rate} &= 1 \text{ lpm/m}^2 \text{ of tank area.} \\
 \text{Cooling water required} &= \pi \times 30 \text{ m} \times 14.4 \text{ m} \times 1 \text{ lpm/m}^2. \\
 &= 1357.7 \text{ lpm.} \\
 &= \frac{1357.7 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= \mathbf{81 \text{ m}^3/\text{hr}}
 \end{aligned}$$

Total Water required for cooling of tanks (item i + ii) = 244+81 = 325 m³/ hr

b) Foam water flow rate

Water flow required for applying foam on a largest tank burning surface area (rim seal area)

For floating roof tank of 30 M diameter,

$$\begin{aligned}
 \text{Diameter of the tank (D1)} &= 30\text{M} \\
 \text{Distance of foam dam from shell} &= 0.8\text{M} \\
 \text{Diameter of roof up to foam dam (D2)} &= 30 - (2 \times 0.8) = 28.4 \\
 \text{Rim seal area} &= (\pi/4) \times (30^2 - 28.4^2) \\
 &= (\pi/4) \times 93.44 \\
 &= 73.4 \text{ m}^2 \\
 \text{Foam solution rate @ 12 lpm/ m}^2 &= 880.8 \text{ lpm} \\
 \text{Foam water required} &= 0.97 \times 880.8 \text{ lpm} \\
 \text{(For 3\% foam concentrate)} &= 854.4 \text{ lpm.} \\
 &= \frac{854.4 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= \mathbf{51 \text{ m}^3/\text{hr.}}
 \end{aligned}$$

Total water flow rate (item a + item b) for floating roof tank protection:

$$\begin{aligned}
 \text{(i) Tank cooling} &= \mathbf{325 \text{ m}^3/\text{hr.}} \\
 \text{(ii) Foam solution application} &= \mathbf{51 \text{ m}^3/\text{hr.}} \\
 \text{Total (item i + ii)} &= \mathbf{376 \text{ m}^3/\text{hr.}}
 \end{aligned}$$

2.2 FIRE WATER FLOW RATE FOR CONE ROOF TANK PROTECTION

Data

$$\begin{aligned}
 \text{Total storage capacity in one dyke area} &= 10,000 \text{ m}^3. \\
 \text{No. of tanks} &= 2 \\
 \text{Capacity of each tank} &= 5000 \text{ m}^3. \\
 \text{Diameter of each tank} &= 24 \text{ m.} \\
 \text{Height of each tank} &= 12 \text{ m.}
 \end{aligned}$$

a) Cooling water flow rate

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(i) Cooling water required for tank on fire

$$\begin{aligned}
 \text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area for tank on fire.} \\
 \text{Cooling water required} &= \pi \times 24 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \\
 &= 2715.4 \text{ lpm.} \\
 &= \frac{2715.4 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= \mathbf{163 \text{ m}^3/\text{hr}}
 \end{aligned}$$

Assuming that other tank is also located within the same tank dyke at a distance less than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 3 lpm/m² of tank shell area.

(ii) Cooling water required for tank falling within (R+30) from centre of tank on fire

$$\begin{aligned}
 \text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\
 \text{Cooling water required} &= \pi \times 24 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \\
 &= 2715.4 \text{ lpm.} \\
 &= \frac{2715.4 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= \mathbf{163 \text{ m}^3/\text{hr}}
 \end{aligned}$$

$$\text{Total cooling water required (item i+ ii)} = \mathbf{163 + 163 = 326 \text{ m}^3/\text{hr.}}$$

b) Foam water flow rate

$$\begin{aligned}
 \text{Foam solution application rate} &= 5 \text{ lpm/m}^2 \text{ of liquid surface area.} \\
 \text{Foam solution required} &= \frac{\pi \times (24 \text{ m})^2 \times 5 \text{ lpm/m}^2}{4} \\
 &= 2262.9 \text{ lpm.} \\
 \text{Foam water required} &= 0.97 \times 2262.9 \text{ lpm} = 2195 \text{ lpm.} \\
 \text{(For 3\% foam concentrate)} &= \frac{2195 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= \mathbf{132 \text{ m}^3/\text{hr.}}
 \end{aligned}$$

Total water flow rate (item a + b) for cone roof tank protection:

$$\begin{aligned}
 \text{(a) Tank cooling} &= \mathbf{326 \text{ m}^3/\text{hr.}} \\
 \text{(b) Foam solution application} &= \mathbf{132 \text{ m}^3/\text{hr.}}
 \end{aligned}$$

$$\text{Total} = \mathbf{458 \text{ m}^3/\text{hr.}}$$

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2.3 FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY

a) Data

Total No. of loading points	=	Conventional or BTPN.
Width of tank wagon gantry (Cooling two spur)	=	12 m.

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time.

Water rate required	=	3 x 15 m x 12 m x 10.2 lpm/m ² .
	=	5508 lpm
	=	330 m³/hr

2.4 Fire water calculation for full surface fire on largest floating roof tank (roof sinking case)

Data:

Total storage capacity in one dyke area	=	20000 m ³
No. of tanks	=	2
Capacity of each tank	=	10,000 m ³
Diameter of each tank	=	30 m
Height of each tank	=	14.4 m

a) Cooling water requirement:

Cooling water rate @ 3 lpm/ m ² of tank shell area for tank-on-fire	
Cooling water required	= $\pi \times 30 \times 14.4 \times 3$
	= 4073.1 lpm
	= 244 m ³ /hr

Assuming that second tank is located within the tank dyke at a distance more than 30M from the tank shell.

Then, cooling water requirement @ 1 lpm/ m ² of tank shell area	= $\pi \times 30 \times 14.4 \times 1$
	= 1357.7 lpm
	= 81 m ³ /hr.

Total cooling water	= (244 + 81) m ³ /hr
	= 325 m³/hr

b) Water requirement in foam application

Foam Application Rate @ 8.1 lpm/m²

Foam Solution Requirement	= $(\pi \times 30 \text{m} \times 30 \text{m}) / 4 \times 8.1 \text{ lpm/m}^2$
	= 5727.9 lpm
	= 344 m ³ /hr



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Water required for the foam solution $= 0.97 \times 344 \text{ m}^3/\text{hr}$
 $= 334 \text{ m}^3/\text{hr}$ refer Note-2

Total water required for roof sink case:

Tank cooling $325 \text{ m}^3/\text{hr}$
 Foam application $334 \text{ m}^3/\text{hr}$ (Plus requirement for foam losses as per Note-2)
Total $659 \text{ m}^3/\text{hr}$
 Total water requirement = **$659 \text{ m}^3/\text{hr}$** (Plus requirement for foam losses as per Note-2)

Note-2: Potential foam losses from wind and other sources to be added to this value as per design requirements. These losses are not considered in this typical calculation sheet.

Note:

Full surface fire of floating roof tank roof sinking case being a remote possibility, it is considered as a single largest contingency for the purpose of arriving at design fire water requirement.

2.5 TOTAL DESIGN FIRE WATER FLOW RATE FOR SINGLE FIRE CONTEGENCY

The total fire water flow requirement will be highest of one of the fire water requirement calculated in 2.1 ($376 \text{ m}^3/\text{hr}$) & 2.2 ($458 \text{ m}^3/\text{hr}$), 2.3 ($330 \text{ m}^3/\text{hr}$) and 2.4 ($659 \text{ m}^3/\text{hr}$) above i.e. $659 \text{ m}^3/\text{hr}$ plus supplementary water ($36 \times 4 = 144 \text{ m}^3/\text{hr}$) = $803 \text{ m}^3/\text{hr}$.

WATER STORAGE REQUIREMENT**Case 1: When make-up water is not available:**

Let us assume two main pumps of capacity $410 \text{ m}^3/\text{hr}$ each and one stand-by pump of equal capacity and equal head are provided.

Water requirement is $410 \times 2 = 820 \text{ m}^3/\text{hr}$
 Design flow rate (Fire water pump discharge) = $820 \text{ m}^3/\text{hr}$

Fire water storage required (4 hrs) $= 820 \times 4$
 $= 3280 \text{ m}^3$

Case-2: When 50% or more make up water is available

Fire water storage requirement (3 hrs) $= 820 \times 3$
 $= 2460 \text{ m}^3$

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3.0 FIRE WATER DEMAND FOR TWO MAJOR FIRES SIMULTANEOUSLY (For locations with aggregate storage capacity more than 30000KL)

Consider various areas under fire and calculate fire water demand for each area based on design basis as indicated below, however, actual tank dimensions available in the terminal shall be considered.

3.1 FIRE WATER FLOW RATE FOR FLOATING ROOF TANK PROTECTION

Data

Total storage capacity in one dyke area	=	32,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	16,000 m ³ .
Diameter of each tank	=	40 m.
Height of each tank	=	14.4 m.

a) Cooling water flow rate

(i) Cooling water required for tank on fire

Cooling water rate	=	3 lpm/m ² of tank area for tank on fire.
Cooling water required	=	$\pi \times 40 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2$.
	=	5430.8 lpm.
	=	$\frac{5430.8 \times 60}{1000} \text{ m}^3/\text{hr}$
	=	326 m³/hr

Assuming that second tank is also located within the same tank dyke at a distance more than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 1 lpm/m² of tank shell area.

(ii) Cooling water required for tank falling beyond (R+30) from centre of tank on fire

Cooling water rate	=	1 lpm/m ² of tank area.
Cooling water required	=	$\pi \times 40 \text{ m} \times 14.4 \text{ m} \times 1 \text{ lpm/m}^2$.
	=	1810.3 lpm.
	=	$\frac{1810.3 \times 60}{1000} \text{ m}^3/\text{hr}$
	=	109 m³/hr.

**Total fire water requirement for cooling of tanks (item i + ii) = 326 + 109
= 435 m³/ hr**

b) Foam water flow rate

Water flow required for applying foam on a largest tank burning surface area (rim seal area)

For floating roof tank of 40 M diameter,

Diameter of the tank (D1) = 40M

Distance of foam dam from shell = 0.8M

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$$\begin{aligned}
 \text{Diameter of roof up to foam dam (D2)} &= 40 - (2 \times 0.8) = 38.4 \\
 \text{Rim seal area} &= (\pi / 4) \times (40^2 - 38.4^2) \\
 &= (\pi / 4) \times 125.44 \\
 &= 98.6 \text{ m}^2 \\
 \text{Foam solution rate @ 12 lpm/ m}^2 &= 1183.2 \text{ lpm} \\
 \text{Foam water required} &= 0.97 \times 1183.2 \text{ lpm} \\
 \text{(For 3\% foam concentrate)} &= 1147.7 \text{ lpm.} \\
 &= \frac{1147.7 \times 60}{1000} \text{ m}^3/\text{hr} \\
 &= 69 \text{ m}^3/\text{hr.}
 \end{aligned}$$

Total water flow rate (item a + b) for floating roof tank protection

$$\begin{aligned}
 \text{Tank cooling} &= 435 \text{ m}^3/\text{hr.} \\
 \text{Foam solution application} &= 69 \text{ m}^3/\text{hr.} \\
 \text{Total water requirement} &= 504 \text{ m}^3/\text{hr.}
 \end{aligned}$$

3.2 FIRE WATER FLOW RATE FOR CONE ROOF TANK PROTECTION

Data

$$\begin{aligned}
 \text{Total storage capacity in one dyke area} &= 50,000 \text{ m}^3. \\
 \text{No. of tanks} &= 4. \\
 \text{Capacity of each tank} &= 12,500 \text{ m}^3. \\
 \text{Diameter of each tank} &= 37.5 \text{ m.} \\
 \text{Height of each tank} &= 12 \text{ m.}
 \end{aligned}$$

a) Cooling water flow rate

(i) Cooling water required for tank on fire

$$\begin{aligned}
 \text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area for tank on fire.} \\
 \text{Cooling water required} &= \pi \times 37.5 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2. \\
 &= 4242.8 \text{ lpm.} \\
 &= \frac{4242.8 \times 60}{1000} \text{ m}^3/\text{hr} = 255 \text{ m}^3/\text{hr.}
 \end{aligned}$$

Assuming that other three tanks are also located within the same tank dyke at a distance less than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 3 lpm/m² of tank shell area.

(ii) Cooling water required for tanks falling within (R+30) from centre of tank on fire

$$\begin{aligned}
 \text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\
 \text{Cooling water required} &= \pi \times 37.5 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \times 3 \\
 &= 12728.6 \text{ lpm.} \\
 &= \frac{12728.6 \times 60}{1000} \text{ m}^3/\text{hr} = 764 \text{ m}^3/\text{hr.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total cooling water required (item i + ii) for cone roof tank protection:} \\
 &= (255 + 764) \text{ m}^3/\text{hr} \\
 &= 1019 \text{ m}^3/\text{hr}
 \end{aligned}$$

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b) Foam water flow rate

Foam solution application rate	=	5 lpm/m ² of liquid surface area.
Foam solution required	=	$\pi \times (18.75 \text{ m})^2 \times 5 \text{ lpm/m}^2$.
	=	5524.5 lpm.
Foam water required	=	$0.97 \times 5524.5 \text{ lpm} = 5358.7 \text{ lpm}$.
(For 3% foam concentrate)	=	$\frac{5358.7 \times 60}{1000} \text{ m}^3/\text{hr}$
Total Foam water required	=	322 m³/hr.

Total water flow rate (item a + b) for cone roof tank protection

Tank cooling	=	1019 m ³ /hr.
Foam solution application	=	322 m ³ /hr.
Total	=	1341 m³/hr.

3.3 FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY
a) Data

Total No. of loading points	=	Conventional or BTPN.
Width of tank wagon gantry (Cooling two spur)	=	12 m.

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time.

Water rate required	=	$3 \times 15 \text{ m} \times 12 \text{ m} \times 10.2 \text{ lpm/m}^2$.
	=	5508
	=	330 m³/hr

3.4 Fire water calculation for full surface fire on largest floating roof tank (roof sinking case)
Data:

Total storage capacity in one dyke area	=	32,000 m ³
No. of tanks	=	2
Capacity of each tank	=	16,000 m ³
Diameter of each tank	=	40 m
Height of each tank	=	14.4 m

a) Cooling water requirement:

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Cooling water rate @ 3 lpm/ m² of tank shell area for tank-on-fire

$$\begin{aligned}
 \text{Cooling water required} &= \pi \times 40 \times 14.4 \times 3 \\
 &= 5430.9 \text{ lpm} \\
 &= 326 \text{ m}^3/\text{hr}
 \end{aligned}$$

Assuming that second tank is located within the tank dyke at a distance more than 30M from the tank shell.

$$\begin{aligned}
 \text{Then, cooling water requirement @ 1 lpm/ m}^2 \text{ of tank shell area} &= \pi \times 40 \times 14.4 \times 1 \\
 &= 1810.3 \text{ lpm} \\
 &= 109 \text{ m}^3/\text{hr}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Total cooling water} &= (326 + 109) \text{ m}^3/\text{hr} \\
 &= 435 \text{ m}^3/\text{hr}
 \end{aligned}$$

b) Water requirement in foam application

Foam Application Rate @ 8.1 lpm/m²

$$\begin{aligned}
 \text{Foam Solution Requirement} &= (\pi \times 40 \text{ m} \times 40 \text{ m}) / 4 \times 8.1 \text{ lpm/m}^2 \\
 &= 10182.9 \text{ lpm} \\
 &= 610.9 \text{ m}^3/\text{hr}
 \end{aligned}$$

$$\begin{aligned}
 \text{Water required for the foam solution} &= 0.97 \times 610.9 \text{ m}^3/\text{hr} \\
 &= 593 \text{ m}^3/\text{hr} \quad \text{..... refer Note-2}
 \end{aligned}$$

Total water required for roof sink case:

$$\begin{aligned}
 \text{Tank cooling} &435 \text{ m}^3/\text{hr} \\
 \text{Foam application} &593 \text{ m}^3/\text{hr} \quad (\text{Plus requirement for foam losses as per Note-2}) \\
 \text{Total} &1028 \text{ m}^3/\text{hr} \\
 \text{Total water requirement} &= 1028 \text{ m}^3/\text{hr} \quad (\text{Plus requirement for foam losses as per Note-2})
 \end{aligned}$$

Note:

Full surface fire of floating roof tank roof sinking case being a remote possibility, it is considered as a single largest contingency for the purpose of arriving at design fire water requirement.

3.5 TOTAL DESIGN FIRE WATER FLOW RATE FOR TWO SIMULTANEOUS FIRE SCENARIO

The total fire water flow requirement will be sum of the two largest fire water requirement calculated in 3.1 (504 m³/hr) & 3.2 (1341 m³/hr), 3.3 (330 m³/hr) above i.e. (1341+504) = 1845 m³/hr plus supplementary water (36 x 4 = 144 m³/hr) = **1989 m³/hr**.

OR water requirement alone as calculated in 3.4 i.e. 1028 m³/hr plus supplementary water (36 x 4 = 144 m³/hr) = 1172 m³/hr

Hence water requirement for double contingency locations shall be 1989 m³/hr

WATER STORAGE REQUIREMENT

Case 1: When make water is not available:

Let us assume three main pumps of capacity 750 m³/hr each and two stand-by pump of equal capacity and equal head are provided.

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Water requirement is $750 \times 3 = 2250 \text{ m}^3/\text{hr}$

Design flow rate (Fire water pump discharge) = $2250 \text{ m}^3/\text{hr}$

Fire water storage required (4 hrs) = 2250×4
= **9000 m^3**

Case-2: When 50% or more make up water is available (consider single largest fire plus supplementary firing)

Fire water storage requirement (3 hrs) = $(1341+144) \times 3 = 1485 \times 3$
= **4455 m^3**
= **5000 m^3 (Say)**

Note-2: Potential foam losses from wind and other sources to be added to this value as per design requirements. These losses are not considered in this typical calculation sheet.

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ANNEXURE - II**FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY****a) Data**

Total No. of loading points	=	Conventional or BTPN.
Width of tank wagon gantry (Cooling two spur)	=	12 m.

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time

Water rate required	=	3 x 15 m x 12 m x 10.2 lpm/m ² .
	=	5508 lpm
	=	330 m³/hr

a) Water Requirement for supplementary Hose:

Water for 4 single hydrant streams	=	4 x 36 = 144 m ³ /hr.
Total water requirement	=	144 m³ / hr.

Total water flow rate for gantry protection

(a) Gantry cooling	=	330 m ³ /hr.
(b) Supplementary hose requirement	=	144m ³ / hr
Total	=	474m³/hr.

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ANNEXURE - III

SAMPLE CALCULATION OF FOAM COMPOUND REQUIREMENT FOR A DEPOT/ TERMINAL

1.0 FOAM COMPOUND CALCULATION FOR SINGLE FIRE CONTIGENCY

1.1 Foam compound calculation for single largest floating roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area	=	20,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	10,000 m ³ .
Diameter of each tank	=	30 m.
Height of each tank	=	14.4 m.

Foam compound requirement for tank

Foam solution application rate	=	12 lpm/m ² of rim seal area of tank.
Foam dam height	=	800 mm.
Diameter of the tank (D1)	=	30M
Distance of foam dam from shell	=	0.8M
Diameter of roof up to foam dam (D2)	=	30- (2X0.8) = 28.4
Rim seal area	=	$(\pi / 4) \times (30^2 - 28.4^2)$
	=	$(\pi / 4) \times 250.2$
	=	73.4 m ²
Foam solution rate @ 12 lpm/ m ²	=	880.8 lpm
3% Foam Compound required	=	26.4 lpm
Foam Compound required for 65 mins.	=	(26.4 x 65) litres = 1,716 litres

1.2 Foam compound calculation for single largest cone roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area	=	10,000 m ³ .
No. of tanks	=	2
Capacity of each tank	=	5,000 m ³ .
Diameter of each tank	=	24 m.
Height of each tank	=	12 m.

Foam compound requirement for tank

Foam solution application rate	=	5 lpm/m ² of liquid surface area of tank.
Foam solution required	=	$(\pi \times (12)^2 \times 5)$ lpm = 2262.9 lpm
Foam compound required (3%)	=	0.03 x 2262.9 lpm = 67.9 lpm.
Foam compound required for 65 minutes	=	65 minutes x 67.9 lpm = 4,414 litres.

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1.3 FOAM COMPOUND CALCULATION FOR FLOATING TANK ROOF SINKING SCENARIO

Capacity of Tank	=	10,000 m ³
Diameter of each tank	=	30 m.
Height of each tank	=	14.4 m.
Foam solution requirement	=	$((\pi / 4) \times (30^2) \times 8.1) \text{ lpm}$ = 5728 lpm (1500 GPM approx.)
Nos. of HVLRs (assuming 750 GPM)	=	750 GPM X 2 Nos.
Foam compound required (3%)	=	0.03 x 5728 lpm = 171.84 lpm.
Foam compound required for 65 minutes	=	65 minutes x 171.84 lpm = 11,170 litres.

1.4 FOAM COMPOUND CALCULATION FOR TWO HOSE STREAMS OF FOAM EACH WITH A CAPACITY OF 1140 LPM.

Foam compound requirement for two foam hose streams of 1140 lpm capacity

Foam solution required	=	2 x 1140 lpm.
Foam compound required (3%)	=	0.03 x 2280 lpm = 68.4 lpm.
Foam compound required for 65 minutes	=	65 minutes x 68.4 lpm = 4,446 litres.

1.5 AGGREGATE QUANTITY OF FOAM COMPOUND FOR SINGLE FIRE CONTIGENCY

The aggregate quantity of foam solution shall be largest of the foam requirements calculated in 1.1 (1,716 litres), 1.2 (4,414 litres), 1.3 (11,170 litres), 1.4 (4,446 litres) i.e. **11,170 litres**

Therefore, foam compound to be stored = 11,170 litres

2.0 FOAM COMPOUND CALCULATION FOR TWO MAJOR FIRES SIMULTANEOUSLY (For locations with aggregate storage capacity more than 30000KL)

2.1 Foam compound calculation for single largest floating roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area	=	32,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	16,000 m ³ .
Diameter of each tank	=	40 m.
Height of each tank	=	14.4 m.

Foam compound requirement for tank

Foam solution application rate	=	12 lpm/m ² of rim seal area of tank.
Foam dam height	=	800 mm.
Diameter of the tank (D1)	=	40 m
Distance of foam dam from shell	=	0.8 m
Diameter of roof up to foam dam (D2)	=	40 - (2X0.8) = 38.4 m

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$$\begin{aligned}
 \text{Rim seal area} &= (\pi / 4) \times (40^2 - 38.4^2) \\
 &= 98.6 \text{ m}^2 \\
 \text{Foam solution rate @ 12 lpm/ m}^2 &= 1183.2 \text{ lpm} \\
 \text{3\% Foam Compound required} &= 35.5 \text{ lpm} \\
 \text{Foam Compound required for 65 mins.} &= \mathbf{2308 \text{ litres}}
 \end{aligned}$$

2.2 Foam compound calculation for single largest cone roof tank in a dyke.

Tank Data

$$\begin{aligned}
 \text{Total storage capacity in one dyke area} &= 50,000 \text{ m}^3. \\
 \text{No. of tanks} &= 4. \\
 \text{Capacity of each tank} &= 12,500 \text{ m}^3. \\
 \text{Diameter of each tank} &= 37.5 \text{ m.} \\
 \text{Height of each tank} &= 12 \text{ m.}
 \end{aligned}$$

Foam compound requirement for tank

$$\begin{aligned}
 \text{Foam solution application rate} &= 5 \text{ lpm/m}^2 \text{ of liquid surface area of tank.} \\
 \text{Foam solution required} &= (\pi \times (18.75)^2 \times 5) \text{ lpm} = 5524.6 \\
 \text{Foam compound required (3\%)} &= 0.03 \times 5524.6 \text{ lpm} = 165.7 \text{ lpm.} \\
 \text{Foam compound required for 65 minutes} &= 65 \text{ minutes} \times 165.7 \text{ lpm} = \mathbf{10,771 \text{ litres.}}
 \end{aligned}$$

2.3 In case of double fire, foam will be required to pour in both tanks simultaneously. Foam compound requirement for 65 minutes = (2308 + 10,771) litres = 13,079 litres.

2.4 FOAM COMPOUND CALCULATION FOR ROOF SINKING SCENARIO

Foam application rate of 8.1 lpm/m²

Data:

$$\begin{aligned}
 \text{Total storage capacity in one dyke area} &= 32,000 \text{ m}^3. \\
 \text{No. of tanks} &= 2. \\
 \text{Capacity of each tank} &= 16,000 \text{ m}^3. \\
 \text{Diameter of each tank} &= 40 \text{ m.} \\
 \text{Height of each tank} &= 14.4 \text{ m.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Foam solution requirement} &= ((\pi / 4) \times (40^2) \times 8.1) \text{ lpm} \\
 &= 10183 \text{ lpm (2690 GPM)} \\
 \text{Nos. of HVLRs (assuming 1000 GPM)} &= 1000 \text{ GPM} \times 3 \text{ Nos.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Foam compound required (3\%)} &= (10183 \times 0.03) \text{ lpm} \\
 &= 305.5 \text{ lpm} \\
 \text{Foam compound required for 65 minutes} &= 65 \text{ minutes} \times 305.5 \text{ lpm} = \mathbf{19858 \text{ litres.}} \\
 &\mathbf{\text{Say 20,000 litres}}
 \end{aligned}$$

2.5 FOAM COMPOUND CALCULATION FOR TWO HOSE STREAMS OF FOAM EACH WITH A CAPACITY OF 1140 LPM.

Foam compound requirement for two foam hose streams of 1140 lpm capacity

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Foam solution required = 2 x 1140 lpm.
Foam compound required (3%) = 0.03 x 2280 lpm = 68.4 lpm.
Foam compound required for 65 minutes = 65 minutes x 68.4 lpm = **4446 litres.**

2.6 AGGREGATE QUANTITY OF FOAM COMPOUND FOR TWO FIRES SIMULTANEOUSLY

The aggregate quantity of foam solution shall be largest of the foam requirements calculated in 2.3 (13,079 litres), 2.4 (20,000 litres) or 2.5 (4,446 litres) i.e. **20,000 litres**

Therefore, foam compound to be stored = 20,000 litres

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ANNEXURE-IV

BRIEF DESCRIPTION OF FIRE FIGHTING FOAM

1.0 FIRE FIGHTING FOAM

Fire fighting foam is a homogeneous mass of tiny air or gas filled bubble of low specific gravity, which when applied in correct manner and in sufficient quantity, forms a compact fluid and stable blanket which is capable of floating on the surface of flammable liquids and preventing atmospheric air from reaching the liquid.

2.0 TYPES OF FOAM COMPOUND

Two Types of foams are used for fighting liquid fires:

2.1 CHEMICAL FOAM

When two or more chemicals are added the foam generates due to chemical reaction. The most common ingredients used for chemical foam are sodium bicarbonate and aluminum sulphate with stabilizer. The chemical foam is generally used in fire extinguishers.

2.2 MECHANICAL FOAM

It is produced by mechanically mixing a gas or air to a solution of foam compound (concentrate) in water. Various types of foam concentrates are used for generating foam, depending on the requirement and suitability. Each concentrate has its own advantage and limitations. The brief description of foam concentrates is given below.

3.0 TYPES OF MECHANICAL FOAM

Mechanical foam compound is classified into 3 categories based on its expansion ratio.

3.1 LOW EXPANSION FOAM

Foam expansion ratio can be up to 50 to 1, but usually between 5:1 to 15:1 as typically produced by self aspirating foam branch pipes.

The low expansion foam contains more water and has better resistant to fire. It is suitable for hydrocarbon liquid fires and is widely used in oil refinery, oil platforms, petrochemical and other chemical industries.

3.2 MEDIUM EXPANSION FOAM

Foam expansion ratio vary from 51:1 to 500:1 as typically produced by self aspirating foam branch pipes with nets. This foam has limited use in controlling hydrocarbon liquid fire because of its limitations w. r. t. poor cooling, poor resistant to hot surface/radiant heat, etc.

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3.4 HIGH EXPANSION FOAM

Foam expansion ratio vary from 501:1 to 1500:1, usually between 750:1 to 1000:1 as typically produced by foam generators with air fans. This foam also has very limited use in controlling hydrocarbon liquid fire because of its limitations w. r. t. poor cooling, poor resistant to hot surface/radiant heat, etc. It is used for protection of hydrocarbon gases stored under cryogenic conditions and for warehouse protection.

4.0 TYPES OF LOW EXPANSION FOAM

4.1 PROTEIN FOAM

The foam concentrate is prepared from hydrolyzed protein either from animal or vegetable source. The suitable stabilizer and preservatives are also added.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The effectiveness of foam is not very good on deep pools or low flash point fuels which have had lengthy preburn time unless applied very gently to the surface.

The concentrate is available for induction rate of 3 to 6%. The shelf life of concentrate is 2 years.

4.2 FLUORO PROTEIN FOAM

This is similar to protein base foam with fluoro-chemical which makes it more effective than protein base foam.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The foam is very effective on deep pools of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is similar to that of protein base foam.

4.3 AQUEOUS FILM FORMING FOAM (AFFF)

The foam concentrate mainly consists of fluoro carbon surfactants, foaming agent and stabilizer. This can be used with fresh water as well as with sea water.

It produces very fluid foam, which flows freely on liquid surface. The aqueous film produced suppresses the liquid vapor quickly. The foam has quick fire knock down property and is suitable for liquid hydrocarbon fires. As the foam has poor drainage rate, the effectiveness is limited on deep pool fires of low flash point fuels which have lengthy pre burn time.

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The concentrate is available for induction rate of 3 to 6% and the shelf life is more than 10 years. This can also be used with non aspirating type nozzles.

4.4 MULTIPURPOSE AFFF

Multipurpose AFFF concentrate is synthetic, foaming liquid designed specially for fire protection of water soluble solvents and water insoluble hydrocarbon liquids. This can be used either with fresh water or sea water.

When applied it forms foam with a cohesive polymeric layer on liquid surface, which suppresses the vapor and extinguishes the fire. The foam is also suitable for deep pool fires because of superior drainage rate and more resistive to hot fuels/radiant heat.

The 3% induction rate is suitable for liquid hydrocarbon fires and 5% for water miscible solvents. The shelf life of concentrate is not less than 10 years. This can also be used with non aspirating type nozzles.

4.5 FILM FORMING FLOURO PROTEIN FOAM (FFFPF)

FFFPF combines the rapid fire knock down quality of conventional film forming AFFF with the high level of post fire security and burn back resistance of flouro-protein foam. The concentrate can either be used with fresh water or sea water.

The foam is suitable for hydrocarbon liquid fires including deep pool fires of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is 5 years. This can also be used with non aspirating type nozzles.

5.0 TYPES OF MEDIUM AND HIGH EXPANSION FOAM

Synthetic foam concentrate is used with suitable devices to produce medium and high expansion foams. This can be used on hydrocarbon fuels with low boiling point. The foam is very light in weight and gives poor cooling effect in comparison to low expansion foams. The foam is susceptible to easy break down by hot fuel layers and radiant heat.

The induction rate in water should vary from 1.5 to 3%. Many of the low expansion foam concentrate can also be used with suitable devices to produce medium / high expansion foam.

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ANNEXURE- V

SYSTEM OF AUTOMATIC ACTUATED RIM SEAL FIRE DETECTION AND EXTINGUISHING SYSTEM FOR EXTERNAL FLOATING ROOF TANKS STORING CLASS- A PETROLEUM

The automatic actuated foam flooding system is a system designed to automatically detect and extinguish the floating roof tank rim seal fire at its incipient stage. The system is mounted on the roof of the tank. The minimum requirement for the design of the system is given below:

1.0 Foam Flooding System

Film Forming Fluro Protein Foam (FFFP) / Aqueous Film Forming Foam (AFFF) type concentrate is used in the system.

1.1 Foam Application System

A large storage tank require one or more than one modular units for foam application in the entire rim seal. Each such unit consists of a foam distribution pipe, laid along the tank perimeter over the rim seal area. The spray nozzles for foam application are mounted on the distribution pipe at suitable intervals. Distribution pipe is permanently connected to a storage vessel containing pre-mix foam and both are placed on the roof. The foam is kept pressurized with nitrogen the premix foam solution is contained in a vessel which is kept charged with nitrogen. The system is designed for minimum foam application rate of @ 18 lpm/ m² of rim seal area. For effective control, foam is discharged in approximately 40 seconds.

1.2 Alarm & Auto Actuation System

In case of fire on the rim seal, it is automatically detected by a device capable to sense the same. The device then actuates the spray system for application of foam in the complete area of rim seal to quickly extinguish the fire in its incipient stage. An audio-visual alarm is also coupled with the detection & extinguishing system for necessary fire alert.

The system includes a fire detector network which senses fire and actuates the automatic release of the extinguishing medium on the rim seal area. Each tank shall have independent detection & extinguishing system.

The validity of the approach must be demonstrated by the designer for an effective total flooding extinguishing system which quickly detects and extinguishes fire in its incipient stage without re-flash. Also, the design considerations should include the impact of the weight of the modules placed on the floating roof.

The detection system needs to be highly reliable and shall work at varied site ambient temperatures for protection of rim seal fire.

The rim seal protection systems shall be of linear hollow metallic tube type detectors with foam based extinguishing media or equivalent system*, for existing/ new as well as for replacements of existing system when due. These detection systems shall be listed and/or

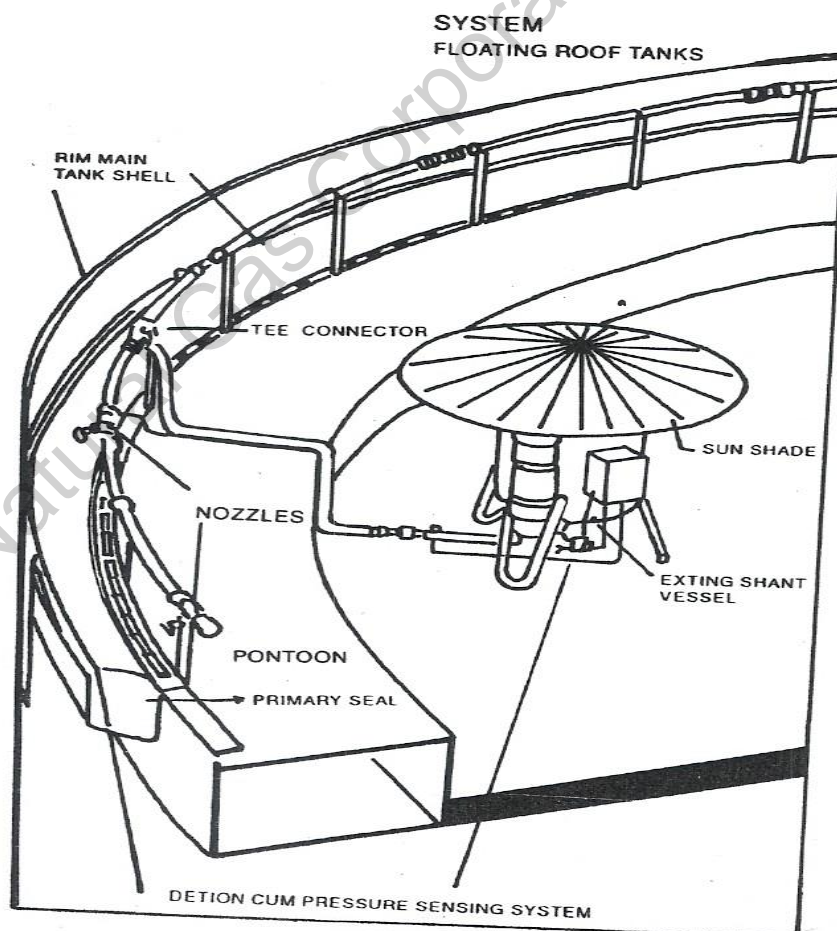


approved by any of the international certifying agencies like UL, FM, VdS or LPC to ensure that those systems are used which meet with highest standards of safety.

* Refer clause at para 2.4 for equivalency.

1.3 Calculations for Modular Foam Application System

Rim seal area of tank = $\pi \times 79 \text{ m} \times 0.3 \text{ m} = 74.5 \text{ m}^2$
 (Considering a flexible seal width of typically 300 mm)
 Foam solution application rate $18 \text{ lpm/m}^2 = 1341 \text{ Liters.}$
 Total foam solution required in 40 seconds = 894 Liters.
 Total nos. of modular unit required = 7 Nos.
 (Considering a vessel of 150 Liters capacity containing 135 Liters of foam)



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Annexure-VI**Explanatory Note for Implementation****Clause 4.2. Design Criteria for fire protection system &****Clause 4.3.2: Basis: Fire water system**

- i. For existing terminals this shall be provided progressively within a period of 4 years from October, 2010.
- ii. For new/upcoming locations it shall be implemented along with commissioning of the facilities.

Clause 4.2.12, 4.4.4

Automatic actuated rim seal fire detection and extinguishing system shall be provided on all external floating roof tanks storing Class A petroleum.

- i. The rim seal protection system shall be implemented in all existing installations progressively within 24 months for class A floating roof tanks above 5000 KL capacity & balance class A floating roof tanks progressively in 36 months from October, 2010.
- ii. Installation of rim seal system shall be done for all new/upcoming external floating roof tanks storing Class A petroleum storage tanks along with commissioning of the tanks.
- iii. The automatic actuated rim seal fire detection & extinguishing system already provided in the existing tanks shall be replaced with a linear hollow metallic tube type detectors with foam based extinguishing media or equivalent system*, whenever the existing system is due for replacement or shall be replaced within a period of 4 years from the day of installation whichever is earlier.

* Refer clause at para 2.4 for equivalency.

Clause: 4.2.13 Sprinkler system for Lube oil drums:

- i. Shall be provided within 12 months from October, 2010.
- ii. For new/upcoming location this shall be implemented along with commissioning of the facilities.

Clause: 4.3.5 (x) Fire Water Pumps

- i. Shall be progressively implemented within 24 months from October, 2010
- ii. For new/upcoming locations it shall be implemented along with commissioning of the facilities.

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Clause 4.3.7 (vi) Hydrants & Monitors:

- i. Shall be provided progressively within 24 months from October, 2010.
- ii. Installation of monitors for all new/upcoming tanks shall be done along with commissioning of the tanks.

Clause 4.4.7: PROTECTION FOR DYKE AREA/SPILL FIRE

- i. Medium expansion foam generators shall be provided progressively within 18 months from October, 2010.
- ii. For new/upcoming location it shall be implemented along with commissioning of the facilities.

Clause 4.6.3 (iii) Accessories:

- i. Emergency Kit with the items mentioned in Annexure-VII shall be provided progressively within 24 months from October, 2010.
- ii. For new/upcoming locations it shall be implemented along with commissioning of the facilities.

Clause: 5.3 DETECTION AND ALARM SYSTEM

- i. Hydrocarbon Detectors shall be provided progressively within 24 months from October, 2010.
- ii. For new/upcoming locations it shall be implemented along with commissioning of the facilities

Clause 6.2 (i) Training:

Training by oil Industry approved reputed institute shall start within next 6 months.

Clause 6.2 (iv) Full activation of shut down system during mock drill:

Shall be implemented with immediate effect.

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Annexure –VII**EMERGENCY KIT**

Emergency Kit consists of listed emergency equipments required for rescue and control/arresting leakage in case of emergency in oil terminals & depots. The equipments shall be mounted on a compact light weight trolley. Emergency Kit shall be consisting of the following emergency equipments:

S.N o.	Item	Quantity	Remarks
1	COLD/LOW TEMPERATURE PROTECTIVE SUIT.	2 sets.	For LPG locations
2	FIRE PROXIMITY SUIT	1 set	
3	PVC SUIT	2 sets	
4	LEAK CONTROL KIT	Consisting of 1 no each of leak arresting pad, leakage control of external pipes, internal pipes, large external pipes up to 8 inch, drums / containers leakages, general purpose leakages, large hole leakages in storage tanks. - 1 set	
5	PETROLEUM PRODUCT CLEANUP CHEMICAL - Boom(5 inch dia , 3 mtr. Long) : 6 nos. - Boom(3 inch dia , 3 mtr. Long) : 6 nos. - Granular particles to absorb Oil : 20 Kg	1 set.	
6	OIL SPILL DISPERSANT (WATER BASED) along with hand held spray nozzle.	Dispersant : 40 litre Spray Gun with back pack : 1 set.	

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S.N o.	Item	Quantity	Remarks
7	NON SPARKING TOOLS One set consisting of : - Shoe handle brush -01 no - 9" Crate opener -01 no - 16oz Claw hammer with Fiberglass handle -01no -Common knife 5 ³ / ₄ " Blade : 1 no 10 ³ / ₄ " OAL, -12" Groove joint plier, -7" Long nose pliers with cutters, -8" Combination Pliers, -Deck scraper, -1 ¹ / ₂ " Blade X 15"Long, -Spray booth scraper, -3" blade X 9 ¹ / ₄ " Long, -Std Screwdriver – ⁵ / ₁₆ " Tip, -6" Blade, - 3" Phillips Screwdriver, -12" Tin Snips, -8" Adjustable Wrench, -12" Adjustable wrench, -14" pipe Wrench (Aluminium), -12" Bung Wrench (Fits ³ / ₄ " X 2")	One Set.	
8	BREATHING APPARTUS SET (40 minute duration)	2 sets	
9	EMERGENCY ESCAPE SETS (15 minutes duration)	2 Sets	
10	FLAME PROOF SEARCH LIGHT	2 nos	Rechargeable type suitable for Explosive Environment.
11	MEGA PHONE EX-PROOF	1 set	Portable battery operated PA System with 1 loud speaker with a range of 1 KM in still air and 500 M in noisy areas.
12	HAND SIREN WITH STAND	1 no	Approx. range of 1.6 KMS
13	FIREMAN AXE	1no	
14	FIRST AID BOX	1 no.	
15	MANUAL RESCUCIATOR	1 no	Manually operated for artificial respirators

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S.N o.	Item	Quantity	Remarks
			consisting of adult size nose, mouth, face plate, air bulb with oxygen inlet connection, non-return, non-breathing human valves and first aid charge packed in a plastic bag
16	FOLDING STRETCHER	1 no	Size 6 feet X 3 feet with tying belts & blanket.
17	MECHANICAL TOOL KIT	1 set	
18	COLD / LOW TEMPERATURE HAND GLOVES	4 Pairs	
19	ELECTRICAL RUBBER HAND GLOVES	2 pair	Suitable for Electrical jobs upto 33000 Volts
20	ELECTRICAL TESTER	1 no	
21	CHEMICAL/OIL SPLASH PROOF GOGGLES	4 nos.	ANSI/CE marked

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